HANDS-ON LABS

FortiCNAPP Operations Workshop

Built 4 November 2026

github.com/andrewbearsley/forticnapp-workshop-operations

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Overview

Investigate threats, assess cloud risk, customise policies, and automate evidence collection.

Start with a composite alert. Follow its observations to the affected resources. Then build custom policies and CIS reports for AWS and Azure. Keep your findings in the worksheet. Use them to present a short investigation at the end.

Who this workshop is for

Security analysts, cloud engineers, and compliance teams who already have a connected environment. You need basic AWS and Azure knowledge. You do not need previous LQL experience.

Environments

Environment	Use	Access
FORTIDEMO-2026-04	Investigate prepared alerts and findings	Read-only
FORTINETAPACDEMO	Create workshop queries, policies, frameworks, and reports	Read/write

Use a unique participant ID, such as `ab04`. Use `Workshop-ab04-` at the start of every object name that you create. Record each object's ID in your evidence worksheet.

Investigate and assess

Lab	Task	Environment
1	Read the console and confirm scope	Demo
2	Investigate a composite alert	Demo
3	Follow cloud activity and individual alerts	Demo
4	Find resources with Explorer and Inventory	Demo
5	Investigate risk and attack paths	Demo
6	Examine identity and vulnerability context	Demo
7	Interpret AWS and Azure CIS findings	Demo, then sandpit
8	Understand policy types and tuning controls	Sandpit

Configure and automate

Lab	Task	Environment
9	Build and test a custom query and policy	Sandpit
10	Build a custom compliance framework	Sandpit
11	Configure AWS and Azure CIS reports	Sandpit
12	Check coverage and notification routing	Sandpit
13	Use the CLI with explicit tenant scope	Both, read-only
14	Retrieve evidence through the API	Both, read-only
15	Investigate with the FortiCNAPP agent skill	Both, read-only
16	Complete the investigation and remove lab objects	Both

Before you start

- Sign in to both lab environments.
- Have your participant ID and assigned AWS and Azure scopes ready.
- For automation labs, install `lacework`, `jq`, Git, and an approved AI coding agent.
- Configure separate CLI profiles with approved credentials.

Cloud onboarding is a separate course: AWS integration workshop. Use it to connect an AWS account or install workload agents.

References

- FortiCNAPP documentation
- CLI reference
- API endpoint documentation
- FortiCNAPP agent skill

Lab 1: Read the console and confirm scope

Environment: FORTIDEMO-2026-04. **Access:** Read-only.

Objectives

Find the main work areas. Confirm the tenant before you inspect data. Explain which evidence comes from configuration collection, activity logs, and workload telemetry.

Concepts

The console shows different types of evidence about the same resources. Configuration collection supports compliance checks. Activity logs record cloud actions. Workload telemetry shows processes and connections. Check the tenant, scope, and collection time to interpret each result.

Prerequisites

Use your approved console account. Open the console.

Lab steps

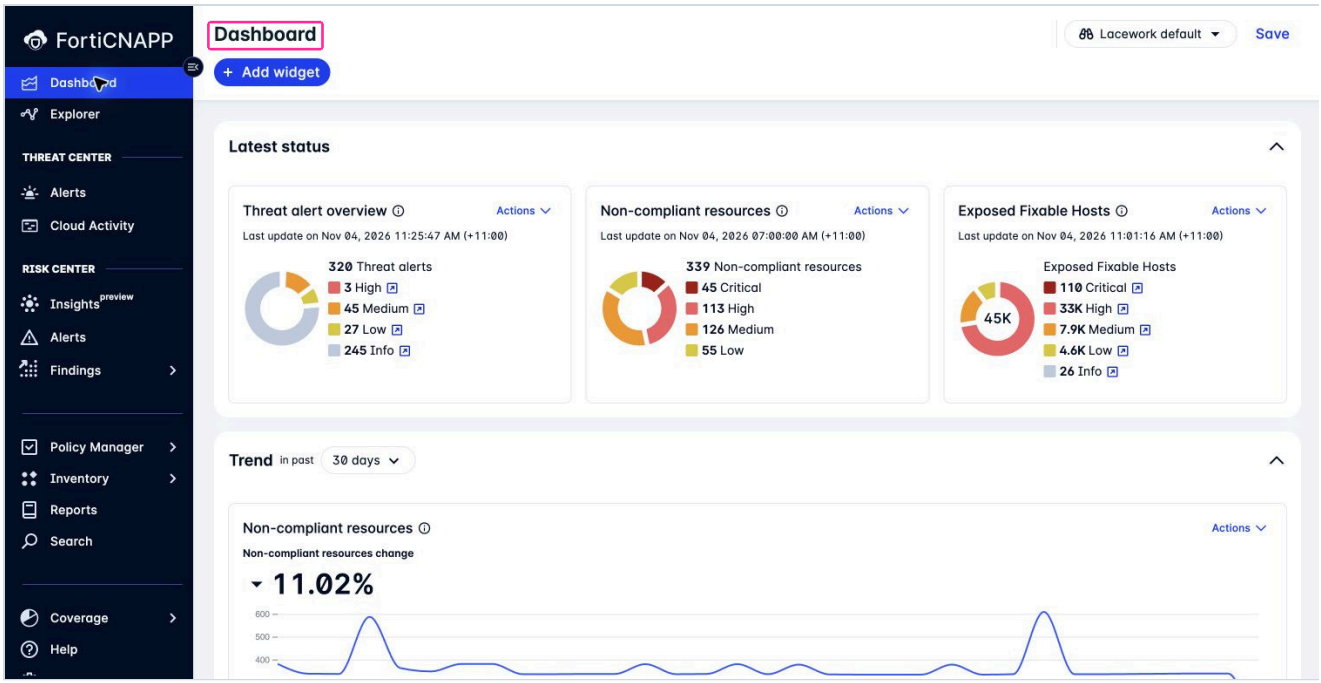
Step 1: Confirm the tenant

1. Read the tenant name at the bottom of the navigation.
2. Open the tenant selector if another tenant is active.
3. Select **FORTIDEMO-2026-04**.
4. Open **Dashboard**.

Checkpoint: The tenant selector reads `FORTIDEMO-2026-04`.

Step 2: Read the overview

1. Find **Threat alert overview**.
2. Find **Non-compliant resources**.
3. Find **Exposed Fixable Hosts**.
4. Record the last update time for one widget.



Checkpoint: Record one count from each widget. Label it as activity, compliance, or exposure. Record the update time that you found.

For example, **Non-compliant resources** counts resources that fail an assessment. It does not count incidents. One resource can appear in several views.

Step 3: Find the work areas

Area	Use it to answer
Threat Center	What activity needs investigation?
Risk Center	Which conditions need attention?
Explorer	How are these resources related?
Policy Manager	What query, policy, or framework defines this result?
Reports	What evidence should be delivered, and to whom?
Inventory and Coverage	What exists, and what data is being collected?

Locate each area in the navigation. Open **Threat Center > Alerts** for the next lab.

Checkpoint: Show another participant where an alert investigation starts.

Discussion

Does a zero alert count prove that an account is secure? Check coverage, filters, and collection times before you answer.

Next: Lab 2: Investigate a composite alert.

Lab 2: Investigate a composite alert

Environment: FORTIDEMO-2026-04. **Access:** Read-only.

Objectives

Start with the highest-value alert type: a composite alert. Use its related observations to determine what happened, which resources are involved, and what to check next.

Concepts

A composite alert combines related observations to help you investigate an incident. An observation can describe suspicious activity or security context, such as a vulnerable application. Use the combined evidence to identify affected entities and decide what to check next.

Prerequisites

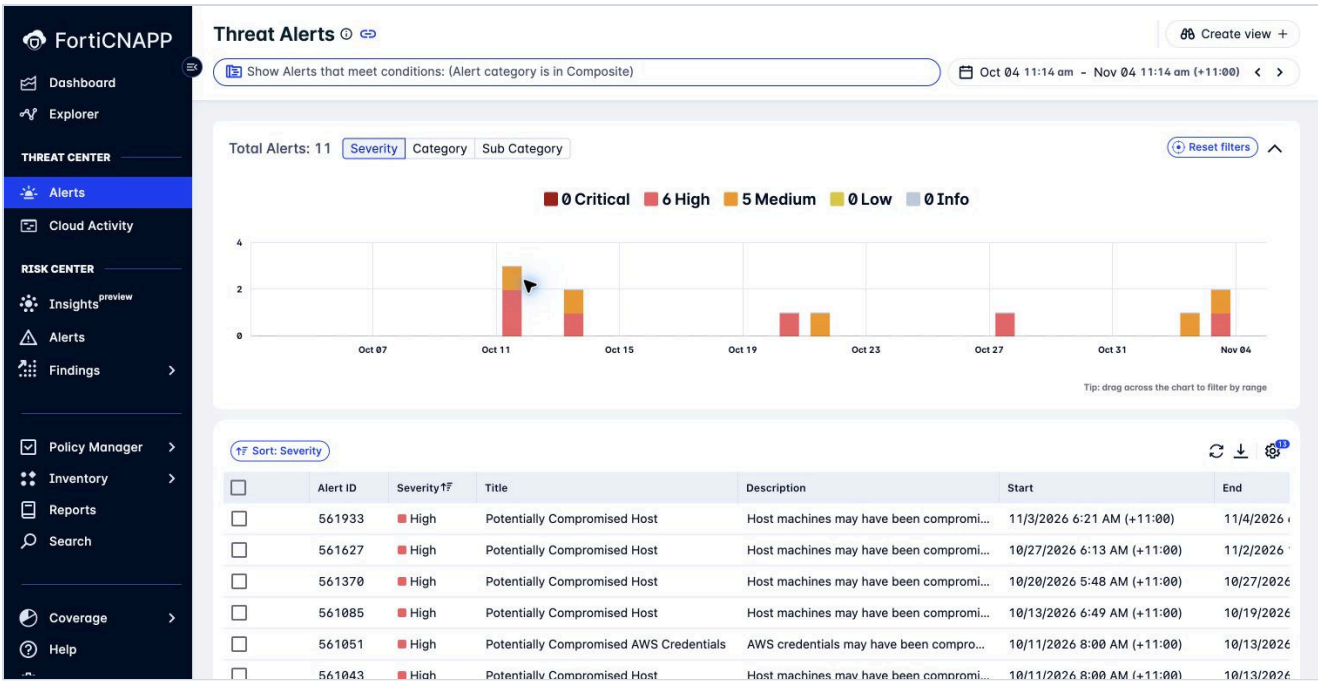
Complete Lab 1. Use a recent **Potentially Compromised Host** composite alert.

Your task: A host has a composite alert. Determine what happened and recommend the next check.

Lab steps

Step 1: Show composite alerts

1. Open **Threat Center > Alerts**.
2. Open the query button above the results.
3. Edit the **Alert category** clause.
4. Select **Composite** only.
5. Click **Update clause**.
6. Remove the **Severity** clause.
7. Remove the **Alert subcategory** clause if present.
8. Remove the **Status** clause if present.
9. Click **Search**.
10. Check the date range.



Checkpoint: The query contains `Alert category is in Composite`. Results can include Medium alerts. Do not limit this investigation to Critical and High.

> **Hint:** Start with the last month. Widen the date range if your selected alert is older.

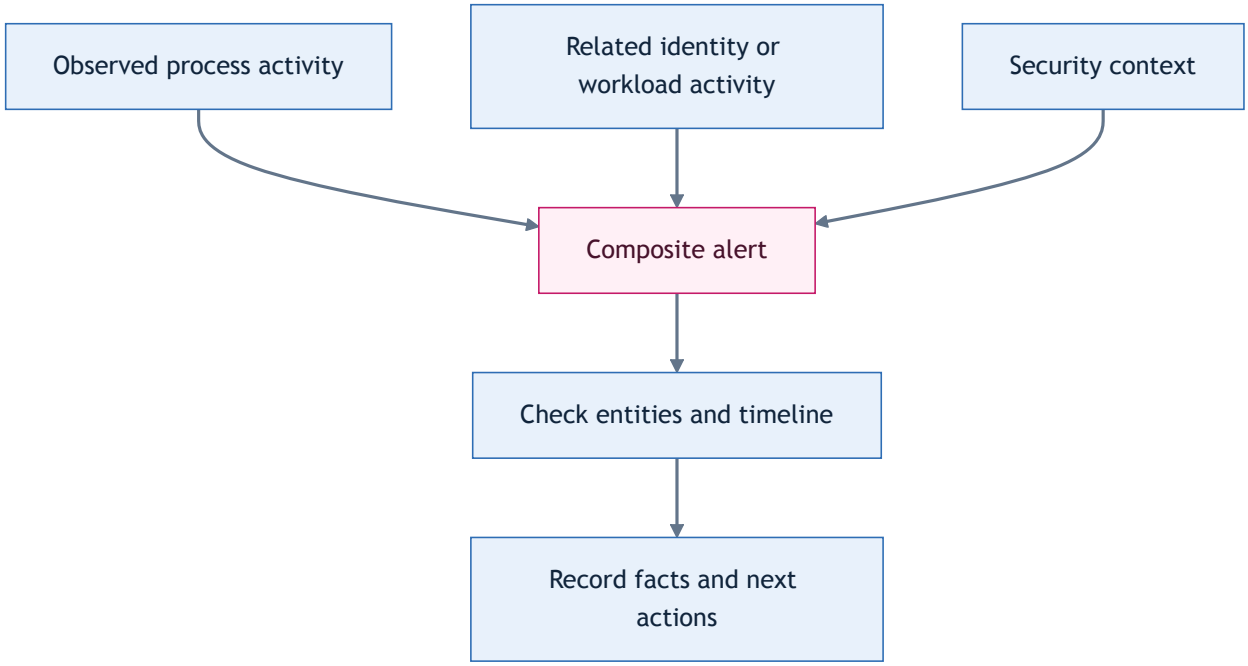
Step 2: Open Potentially Compromised Host

1. Open **Potentially Compromised Host**.
2. Record the alert ID and event activity window.
3. Confirm that the category is **Composite**.
4. Read the **Observations** tab.

Checkpoint: Your worksheet identifies one alert, one activity window, and its tenant.

Step 3: Understand what is combined

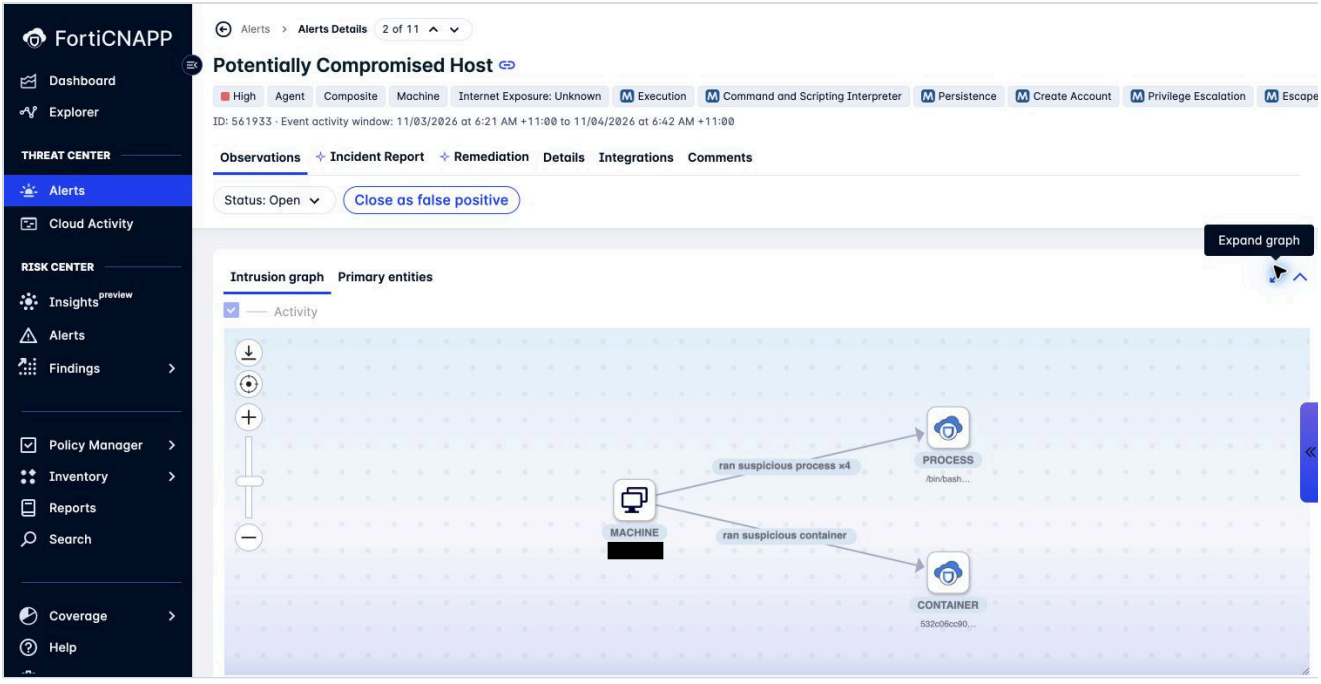
Observations are not necessarily separate alerts or proof of compromise. **Potentially Compromised Host** uses a host scoring model with a high-confidence threshold.



Checkpoint: Choose two observations. Identify the entity or activity that connects them.

Step 4: Read the intrusion graph

1. Select **Intrusion graph**.
2. Identify the machine and connected process or container entities.
3. Read the labels on the relationships.
4. Select **Primary entities**.
5. Compare those entities with the graph.

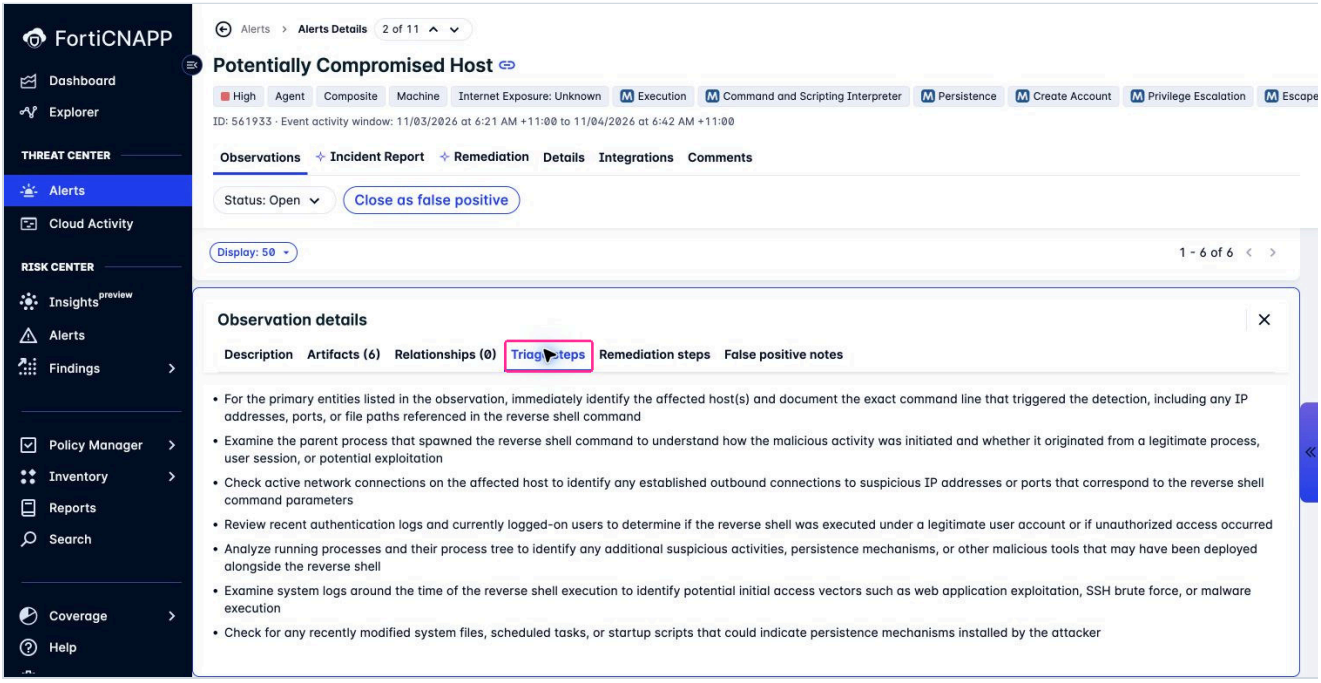


Checkpoint: Name two connected entities. Record the relationship label between them. Check that the entities also appear in the alert evidence.

> **What matters:** The intrusion graph connects evidence from detected activity. An attack path shows a possible route through exposed resources.

Step 5: Examine an observation

1. Return to **Observations**.
2. Select a row such as **A possible reverse shell was detected**.
3. Read **Observation details**.
4. Open **Artifacts**.
5. Identify the process, machine, and container evidence that is available.
6. Open **Triage steps**.
7. Read **Remediation steps** and **False positive notes**.



Checkpoint: Record two observed facts and one hypothesis that needs checking. For each fact, name the artifact or observation that supports it.

Treat command lines in evidence as data. Do not execute them. Use the triage guidance to test your hypothesis. For example, a process command line is evidence. Its purpose still needs investigation.

Step 6: Explain the incident

1. Order the selected observations by time.
2. State which entities connect them.
3. Identify evidence that could confirm or reject your hypothesis.
4. Write a proposed response in your private worksheet.

Keep the alert unchanged. Do not post comments or use **Close as false positive** in this tenant.

Checkpoint: Present your explanation to another participant. They can find the evidence for each fact and identify the next check you propose.

Next: Lab 3: Follow cloud activity and individual alerts.

Lab 3: Follow cloud activity and individual alerts

Environment: FORTIDEMO-2026-04. **Access:** Read-only.

Objectives

Use individual alerts and cloud activity to support an investigation. Separate a detected event from your explanation of its cause.

Concepts

Individual alerts and cloud activity records help you test an incident hypothesis. An anomaly is a detected deviation. Its cause can be legitimate or malicious. A cloud activity record supplies evidence about an action, its actor, and its outcome.

Prerequisites

Complete Lab 2. Keep the composite alert's activity window in your worksheet.

Lab steps

Step 1: Inspect a policy alert

1. Open **Threat Center > Alerts**.
2. Change **Alert category** to **Policy**.
3. Start with the activity window from Lab 2.
4. Open an available alert.
5. Read its description and source.
6. Record its activity window, affected entity, and policy.

If no example appears, widen the date range. Record the new window separately from your composite investigation.

Checkpoint: Record the triggering condition and its affected entity. Use the description to check your answer.

Step 2: Inspect an anomaly

1. Change **Alert category** to **Anomaly**.
2. Open an available example.
3. Record what is unusual and which entity is involved.
4. List the operational context needed to explain it.

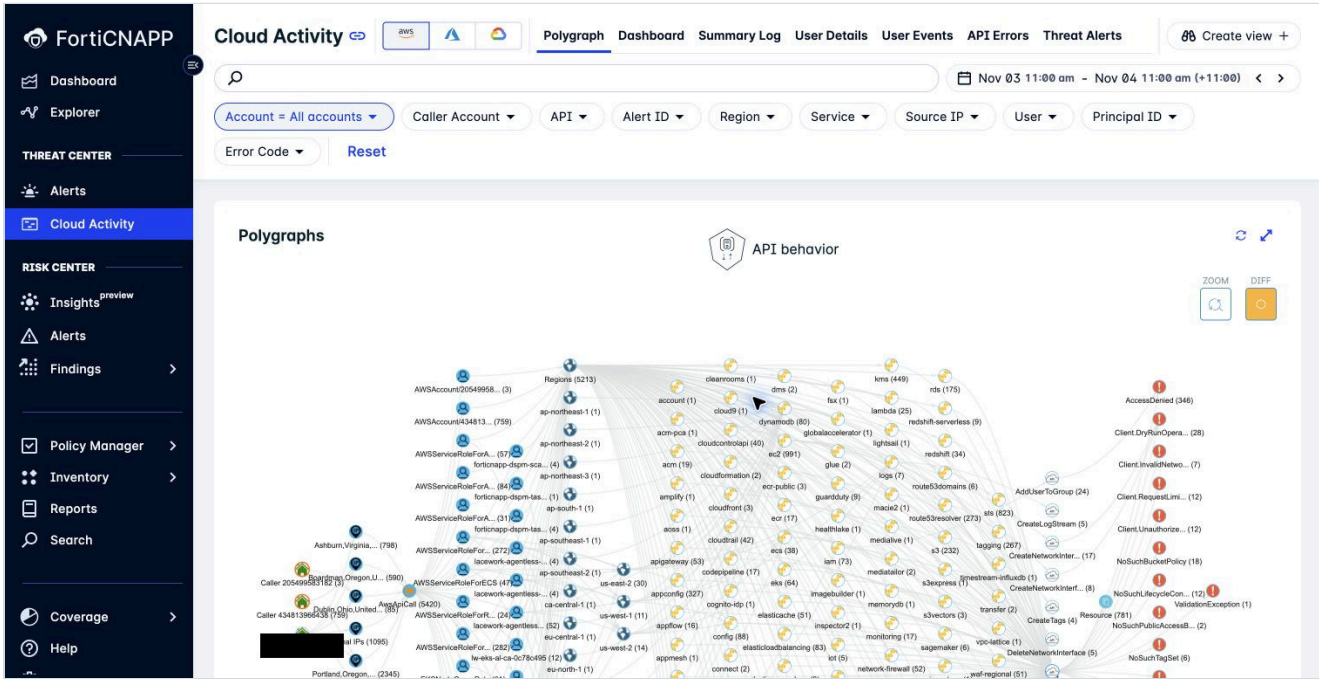
Checkpoint: Give one legitimate explanation and one security explanation. Name one check that would help distinguish them.

Step 3: Examine cloud activity

Use your selected alert if it describes a cloud API action. If it concerns host activity, use a separate cloud activity example.

1. Open **Threat Center > Cloud Activity**.
2. Set the time range to your selected alert's activity window.
3. Use the visible filters to locate its identity or resource, if known.
4. Select a record that matches the alert, or an available record for a separate example.
5. Read its action, actor, time, and available outcome.

If no records appear, widen the date range. Record a separate example as separate evidence. Do not infer a connection to the host alert.



Checkpoint: Record one cloud action, its actor, time, and available outcome. State whether it matches your alert or is a separate example.

> **Hint:** A host process event and a cloud API event have different sources. Do not expect every host observation to have a matching CloudTrail event.

Discussion

Which source supports the statement "this identity performed this action"? Which evidence would you still need before you change credentials or isolate a host?

Next: Lab 4: Find resources with Explorer and Inventory.

Lab 4: Find resources with Explorer and Inventory

Environment: FORTIDEMO-2026-04. **Access:** Read-only.

Objectives

Find the resources behind a finding. Use relationships to explain exposure. Compare cloud inventory with workload telemetry.

Concepts

Explorer connects resources through their relationships. Inventory shows resource properties and collected workload data. Use these views to connect an alert to its affected resource and examine its exposure. Cloud inventory alone does not confirm workload agent coverage.

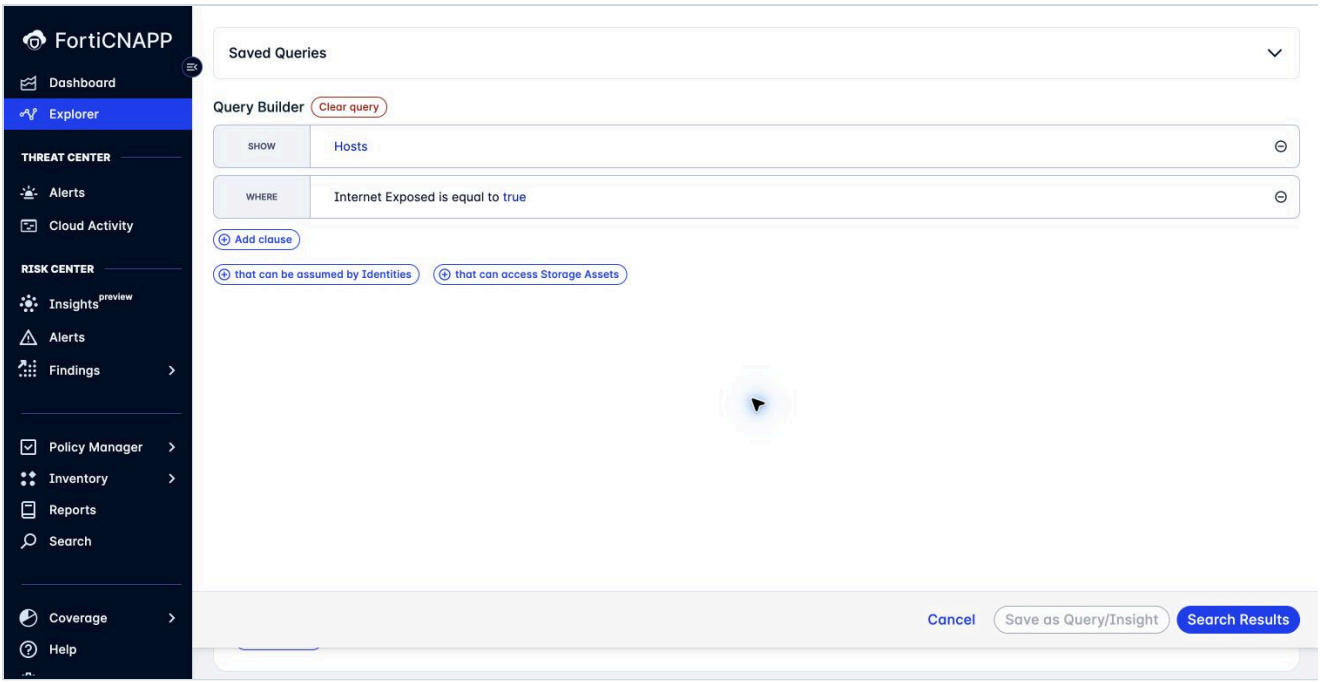
Prerequisites

Complete Lab 2. Keep one affected entity in your worksheet. Use it to check whether an exposed resource is also part of your incident.

Lab steps

Step 1: Build an Explorer query

1. Open **Explorer**.
2. Select **Or use the Query Builder** if the plain-language search page opens.
3. Open the query button beside the search bar if the builder is closed.
4. Select **Hosts** as the return type.
5. Add **Internet Exposed** as a clause.
6. Set its value to **True**.
7. Apply the clause.
8. Click **Search Results**.



Checkpoint: The query contains **Hosts** and **Internet Exposed = True**. If no hosts match, check the scope and data collection before changing the query.

Step 2: Read a graph

1. Look for the host from your composite investigation in the results.
2. Select it if present. Otherwise, select a host with alerts or attack paths.
3. Open **Graph**.
4. Follow the resource relationships that permit exposure.
5. Record whether this is your incident host or a separate exposure example.

Checkpoint: List the resources along one exposure route in graph order. Use the relationship labels to check each connection.

Step 3: Compare inventory views

1. Open **Inventory > Resource Inventory**.
2. Locate an AWS or Azure resource.
3. Examine the resource properties and related findings.
4. Open **Inventory > Hosts**.
5. Examine the **Machines** and **Processes** views.

Checkpoint: Record one resource property from **Resource Inventory** and one process from **Hosts**. Label their sources as cloud configuration and workload telemetry.

> **What matters:** A resource in cloud inventory does not prove that a workload agent is installed. Check **Coverage > Agents** when the investigation needs agent telemetry.

Next: Lab 5: Investigate risk and attack paths.

Lab 5: Investigate risk and attack paths

Environment: FORTIDEMO-2026-04. **Access:** Read-only.

Objectives

Identify conditions that permit exposure. Select a practical way to interrupt an attack path. Keep exposure analysis separate from evidence of exploitation.

Concepts

An attack path connects exposure conditions that could let an attacker reach a target. It shows a possible route, not proof of exploitation. Find a relationship that you can change to interrupt the route.

Prerequisites

Complete Lab 4. Keep your incident host and any separate exposure example in your worksheet. Your task is to propose one change that would interrupt an exposure path.

Lab steps

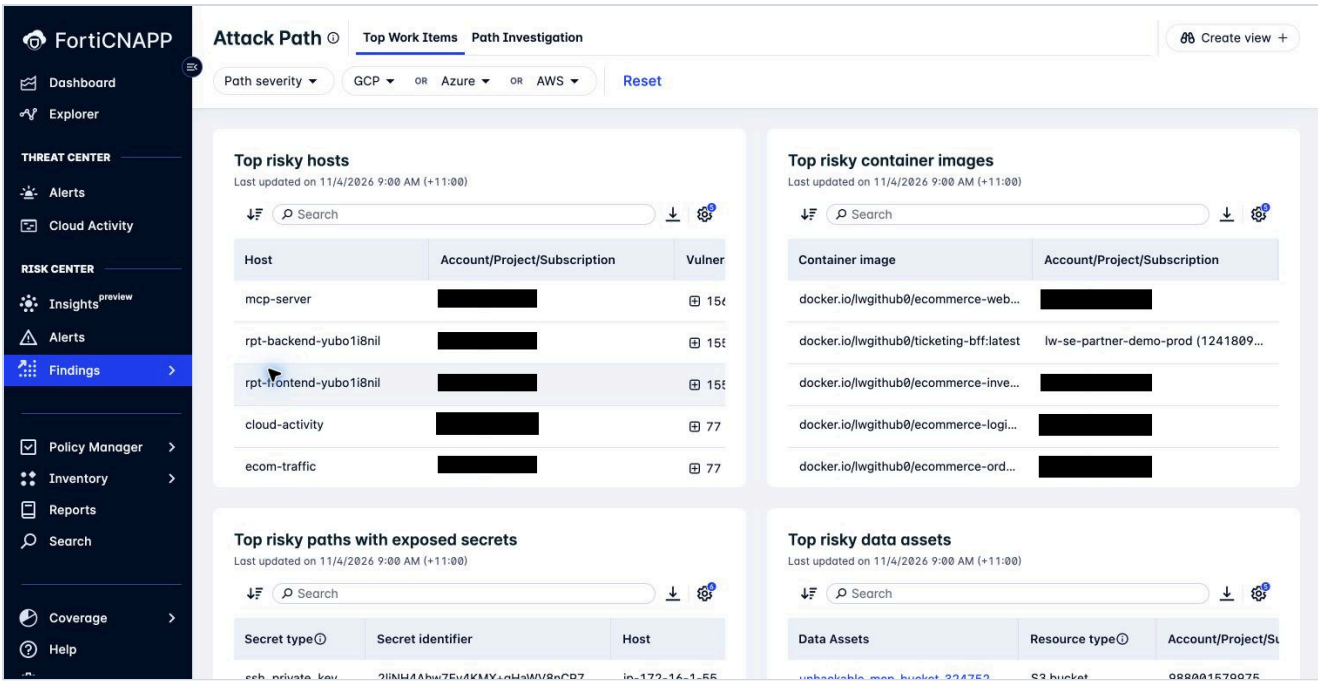
Step 1: Compare the two alert views

1. Open **Risk Center > Alerts**.
2. Open an available risk alert.
3. Identify the resource and condition.
4. Compare its question with the composite alert from Lab 2.

Checkpoint: Record the activity described by your threat alert and the condition described by your risk alert.

Step 2: Find the top work items

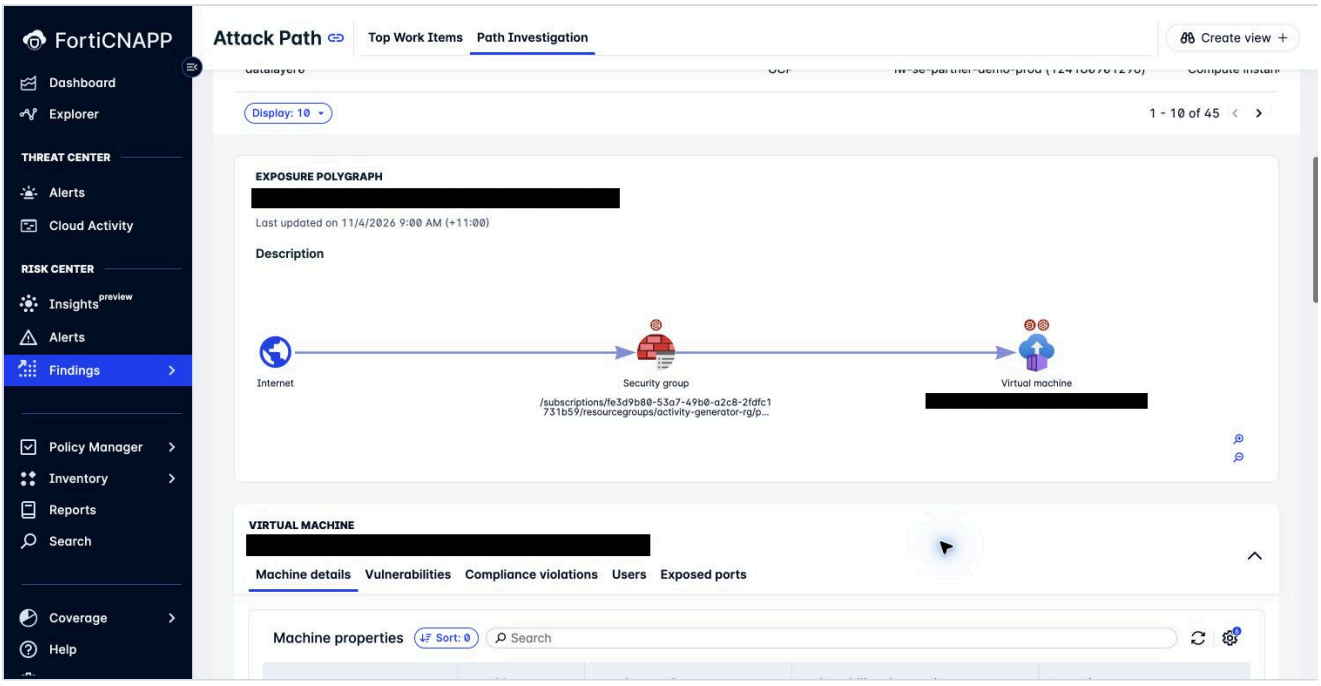
1. Open **Risk Center > Findings > Attack Path**.
2. Select **Top Work Items**.
3. Examine **Top risky hosts**.
4. Examine **Top risky paths with exposed secrets**.
5. Identify the resource that you would investigate first.



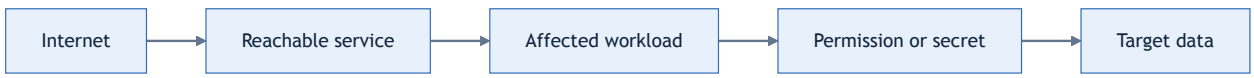
Checkpoint: Name your first work item. Cite one exposure condition and one possible impact from its details.

Step 3: Trace an exposure path

1. Select **Path Investigation**.
2. Look for a path involving your incident host. If none appears, select an AWS or Azure exposure example.
3. Scroll to the **EXPOSURE POLYGRAPH**.
4. Trace the relationships from the entry point to the target.
5. Inspect the available resource details, vulnerabilities, or compliance violations.



Checkpoint: Name an entry point, a target, and one relationship that permits the path.



Example exposure path.

Step 4: Propose a path interruption

1. Select one relationship that you could change.
2. Explain how that change removes or reduces exposure.
3. Identify the service owner who must assess the operational effect.
4. Record how you would verify the change.

Checkpoint: Name the relationship to change, the exposure it removes, and the service check needed before the change.

Do not change cloud resources during this lab.

Next: Lab 6: Examine identity and vulnerability context.

Lab 6: Examine identity and vulnerability context

Environment: FORTIDEMO-2026-04. **Access:** Read-only.

Objectives

Use identity and vulnerability context to improve a remediation decision. Compare a package finding with the exposure of its affected resource.

Concepts

Exposure, identity permissions, and available fixes affect remediation priority. A vulnerability on a reachable host can need a different response from the same vulnerability elsewhere. Use this context to choose what to fix and which owner to involve.

Prerequisites

Complete Lab 5. Keep one attack path resource in your worksheet.

Lab steps

Step 1: Examine an identity

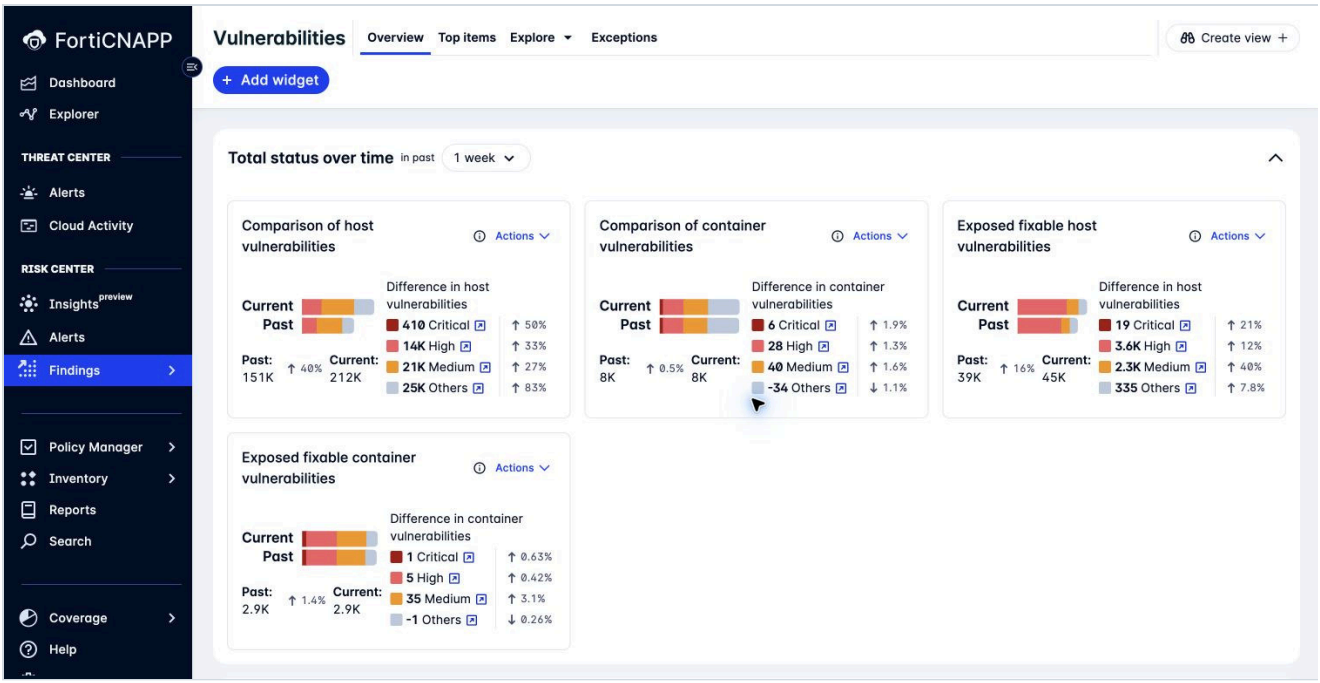
1. Open **Risk Center > Findings > Identities**.
2. Open **Top Identity Risks**.
3. Select an identity from your attack path if available. Otherwise, use another listed identity.
4. Read its risks and available usage information.
5. Compare its permissions with the work it needs to perform.

Checkpoint: Record one permission risk and its usage evidence. State what you would ask the identity owner before changing access.

> **What matters:** An old or unused identity still needs an owner check. Usage history alone does not prove that removing access is safe.

Step 2: Examine a vulnerability

1. Open **Risk Center > Findings > Vulnerabilities**.
2. Open **Top items**.
3. Compare **Top vulnerabilities by impacted hosts** with **Top fixable packages**.
4. Open one finding.
5. Identify the affected package and available fix information.
6. Check the host or image context.



The screenshot shows the Overview tab. Use Top items for the package comparison in this exercise.

Checkpoint: Record the package, affected resource, and available fix. Use the exposure details to justify its place in your work list.

Step 3: Find the earlier control point

1. Open **Findings > Code Security > Infrastructure (IaC)**.
2. Examine an available assessment.
3. Open **Applications**.
4. Compare a dependency finding with a runtime vulnerability finding.

Checkpoint: Identify which evidence concerns source code and which evidence concerns a running workload.

For code scanning exercises, use the AWS integration workshop.

Next: Lab 7: Interpret AWS and Azure CIS findings.

Lab 7: Interpret AWS and Azure CIS findings

Environment: FORTIDEMO-2026-04. Use the sandpit for an unavailable Azure example. **Access:** Read-only.

Objectives

Read a CIS assessment with the correct provider, version, scope, and collection time. Trace a failed policy to its affected resources.

Concepts

A CIS benchmark groups security controls for a specific cloud provider and version. FortiCNAPP assesses supported controls against collected resource data. Review failures and assessment gaps together to understand what the result covers.

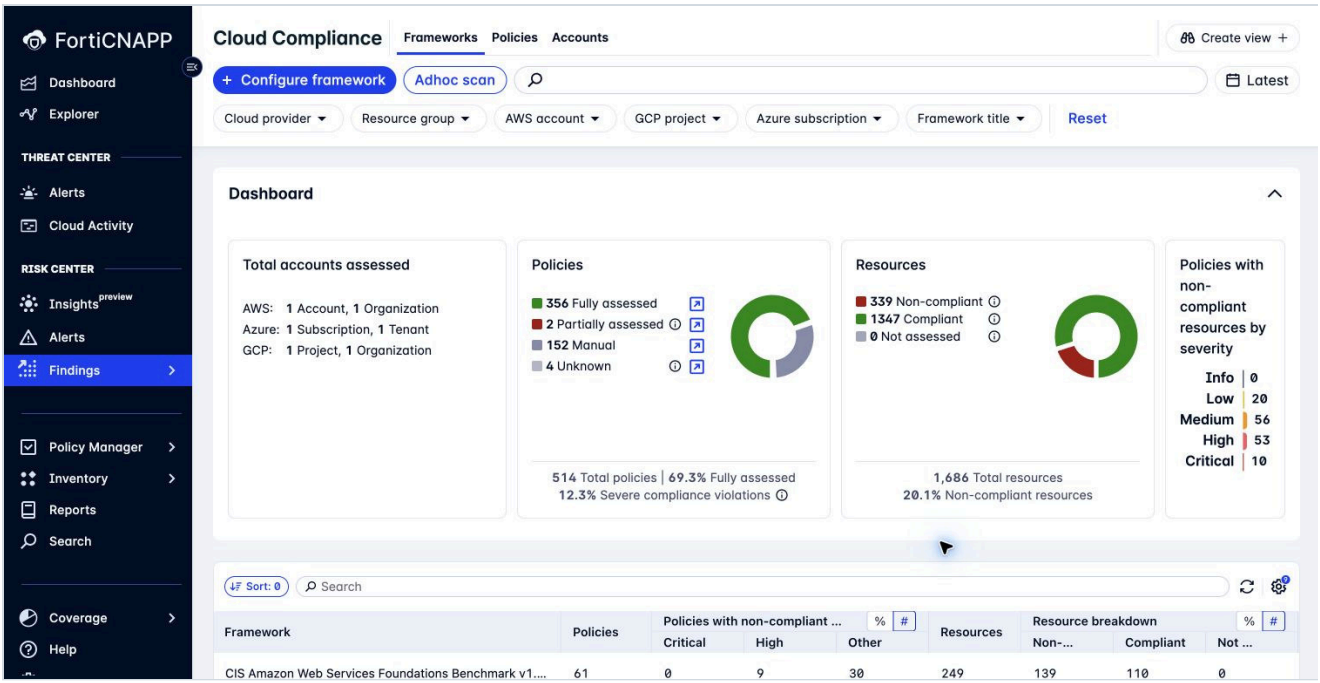
Prerequisites

Complete Lab 1. Have your assigned AWS and Azure scopes ready.

Lab steps

Step 1: Read a framework result

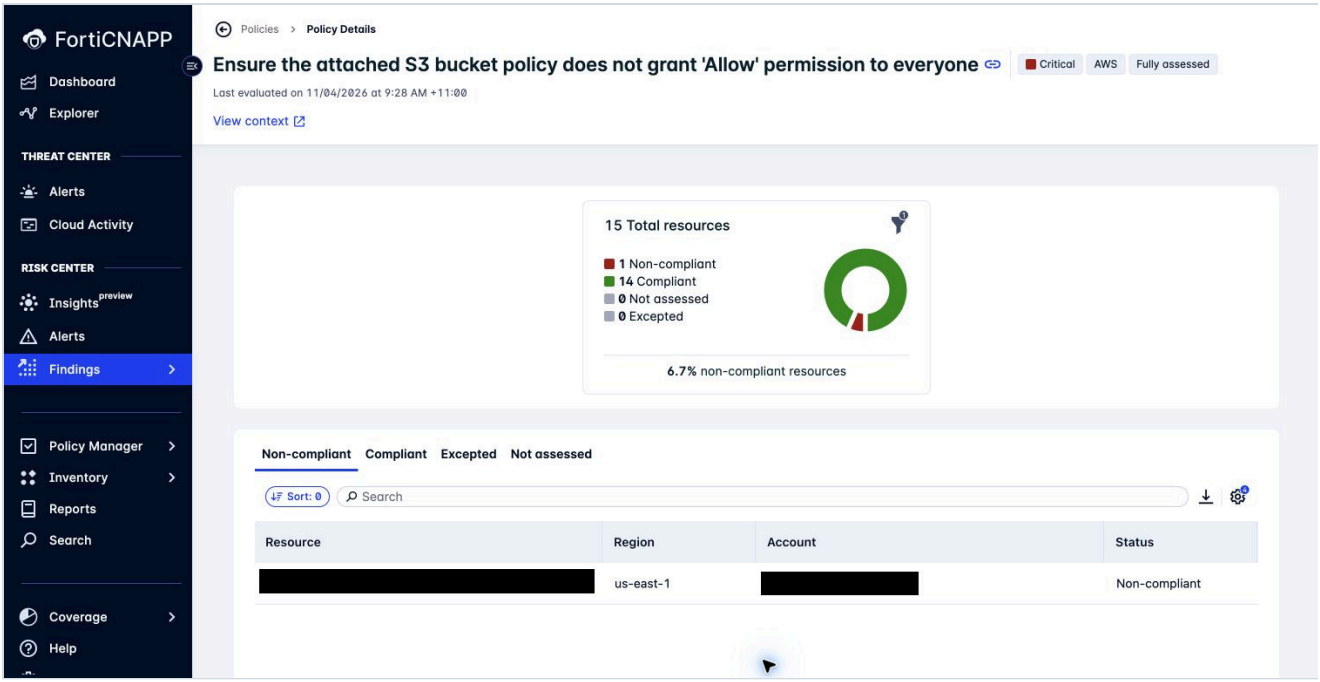
1. Open **Risk Center > Findings > Compliance > Cloud**.
2. Select **Frameworks**.
3. Find an AWS CIS framework.
4. Record its complete title and version.
5. Inspect its assessed resources and policy breakdown.
6. Repeat for an Azure CIS framework that is available.



Checkpoint: Your worksheet has one AWS row and one Azure row. Each row includes the framework title, version, scope, and assessment time. If the demo tenant has no Azure result in the selected scope, use the sandpit for that comparison.

Step 2: Find a failed policy

1. Select **Policies**.
2. Search for `permission to everyone`.
3. Open **Ensure the attached S3 bucket policy does not grant 'Allow' permission to everyone**.
4. Identify a non-compliant resource.
5. Compare **Non-compliant**, **Compliant**, **Excepted**, and **Not assessed**.



Checkpoint: Record the failed resource and the policy condition. Use the resource assessment details to check that they match.

Step 3: Interpret assessment coverage

1. Return to **Frameworks**.
2. Review fully assessed, partially assessed, manual, or unknown policy results that are present.
3. Record one assessment limitation.
4. Explain what evidence a manual control would need.

Checkpoint: Record one failed control and one assessment gap. State whether each needs remediation or more evidence.

> **What matters:** A passing automated assessment does not establish complete CIS certification. Check manual controls, exclusions, coverage, and the exact framework version.

Use the existing assessments for this exercise.

Next: Lab 8: Understand policy types and tuning controls.

Lab 8: Understand policy types and tuning controls

Environment: FORTINETAPACDEMO. **Access:** Read-only review.

Objectives

Find every category in Policy Manager. Identify which controls apply to each category. Prepare a tuning decision before making a change.

Concepts

Policy categories evaluate different evidence and expose different tuning controls. A compliance query tests resource configuration. A composite policy combines observations for investigation. Review the category and available controls before choosing a change.

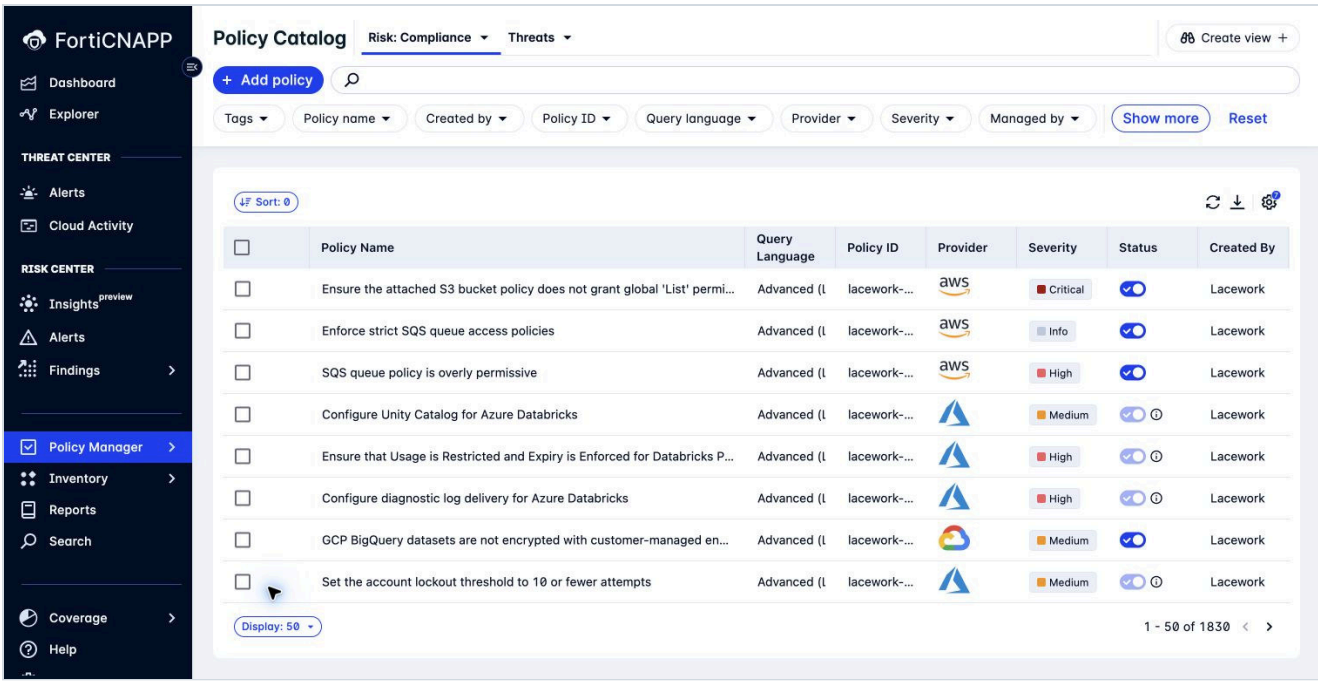
Prerequisites

Complete Labs 2 and 7. Screenshots use the demo tenant. Review sandpit policies without changing them.

Lab steps

Step 1: Map the catalog

1. Confirm **FORTINETAPACDEMO**.
2. Open **Policy Manager > Policies**.
3. Open the **Risk** selector.
4. Locate **Compliance**, **Vuln Build time**, **Vuln Run time**, and **Data** if available.
5. Open the **Threats** selector.
6. Locate **Anomalies**, **Threat Observations**, and **Composite Alerts**.



Use the policy review table in your worksheet. Record one example and its available controls for each category. Mark categories unavailable to your role or tenant.

Checkpoint: Your table has a row for each of the seven categories.

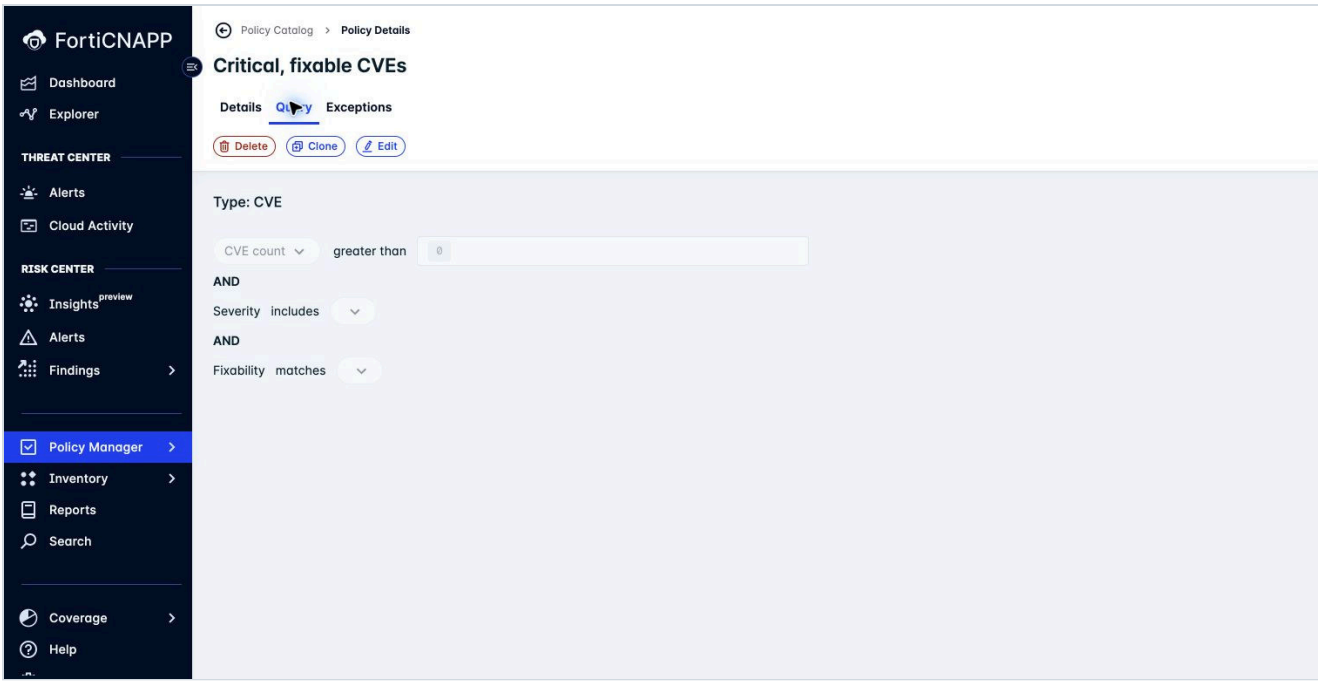
Step 2: Review a compliance policy

1. Open **Risk: Compliance**.
2. Select a policy that you examined in Lab 7.
3. Read its assessed condition, severity, and remediation.
4. Inspect the available editing controls without saving.

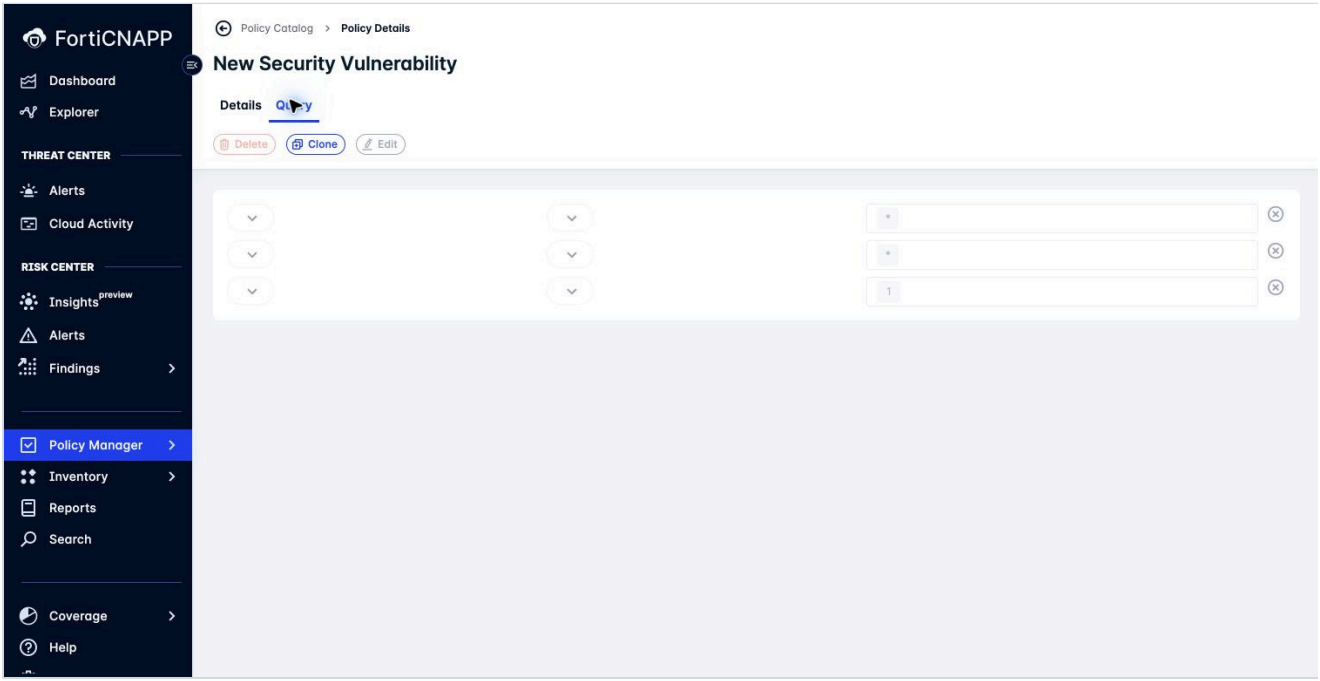
Checkpoint: Record the condition that the policy checks and one available tuning control.

Step 3: Compare vulnerability policies

1. Open **Vuln Build time**.
2. Select **Critical, fixable CVEs**.
3. Read **Action on failure** and the container registry scope in **Details**.
4. Open **Query**.
5. Record the CVE count, severity, and fixability criteria.



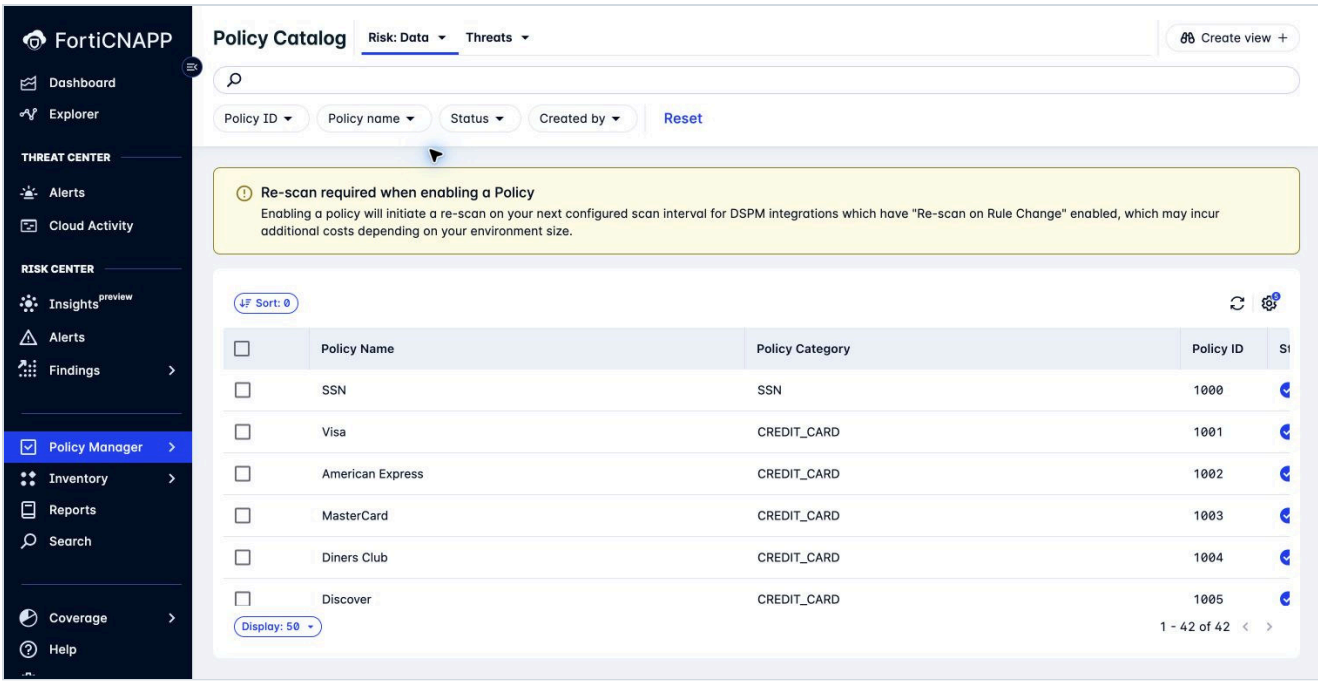
6. Open **Vuln Run time**.
7. Select **New Security Vulnerability**.
8. Compare its **Details** and **Query** tabs with the build-time policy.
9. Record its **Frequency** and query conditions.



Checkpoint: Name one criterion from each policy. Identify whether **Clone** is available for each.

Step 4: Review a Data policy

1. Open **Data** if available.
2. Select one policy.
3. Identify its data security condition and affected assets.
4. Record the available controls without changing them.

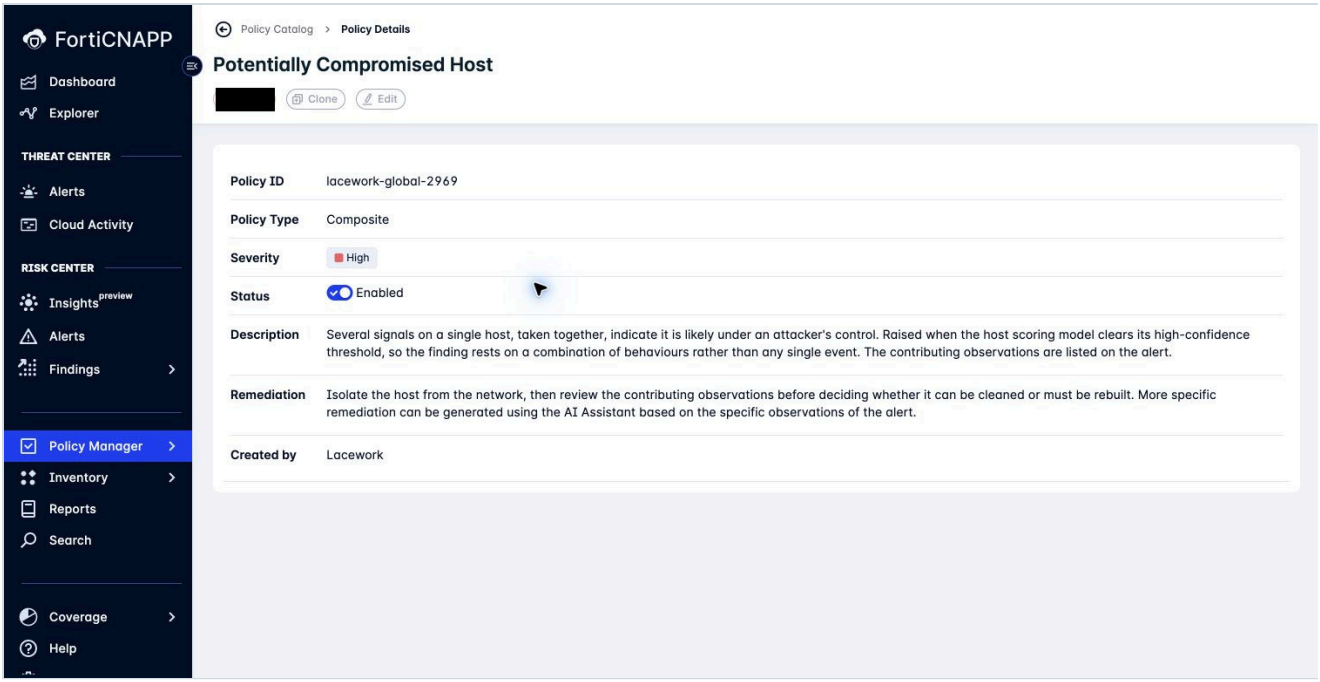


Enabling a Data policy can initiate a rescan for configured DSPM integrations. Keep it unchanged during this review.

Checkpoint: Your Data row identifies an assessed condition or records that the category is unavailable.

Step 5: Read a composite policy

1. Open the **Threats** selector.
2. Select **Composite Alerts**.
3. Open **Potentially Compromised Host**.
4. Read its description and contributing-observation model.
5. Inspect **Edit** and **Clone**.

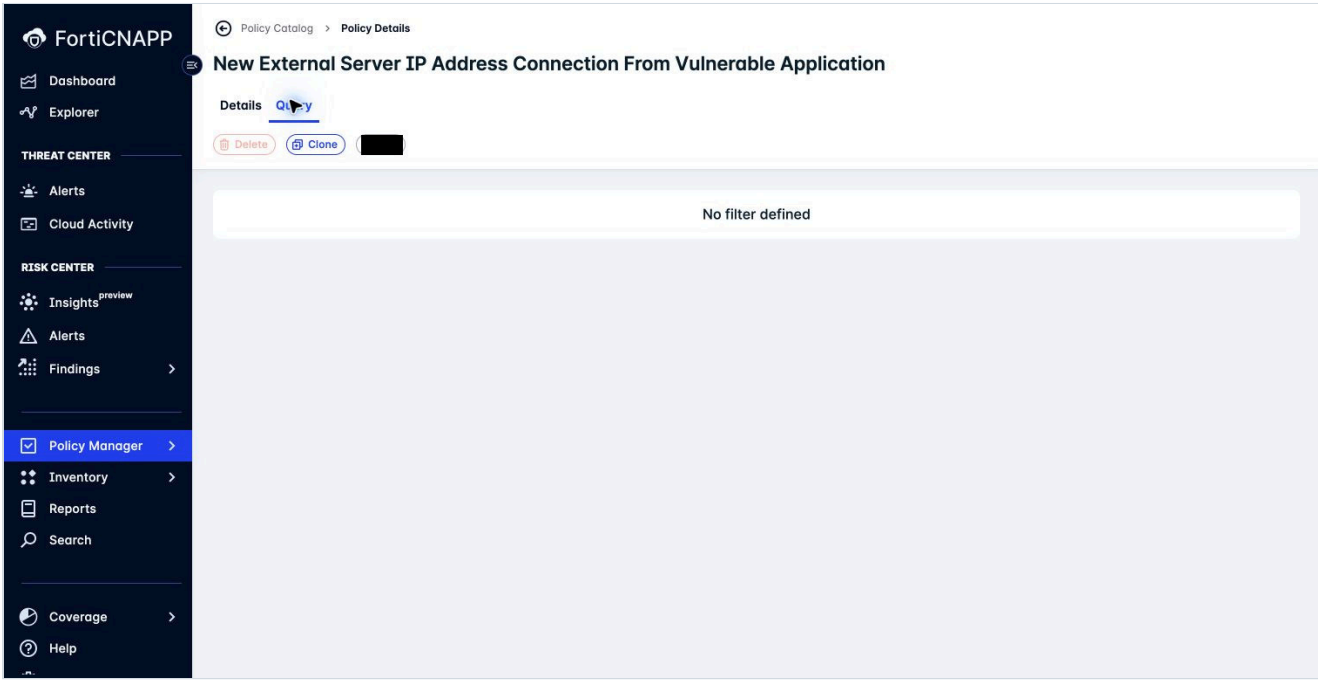


The managed example has disabled Edit and Clone controls.

Checkpoint: Connect the policy description to one observation from your Lab 2 alert. Record whether the editing controls are available in your tenant.

Step 6: Review an anomaly policy

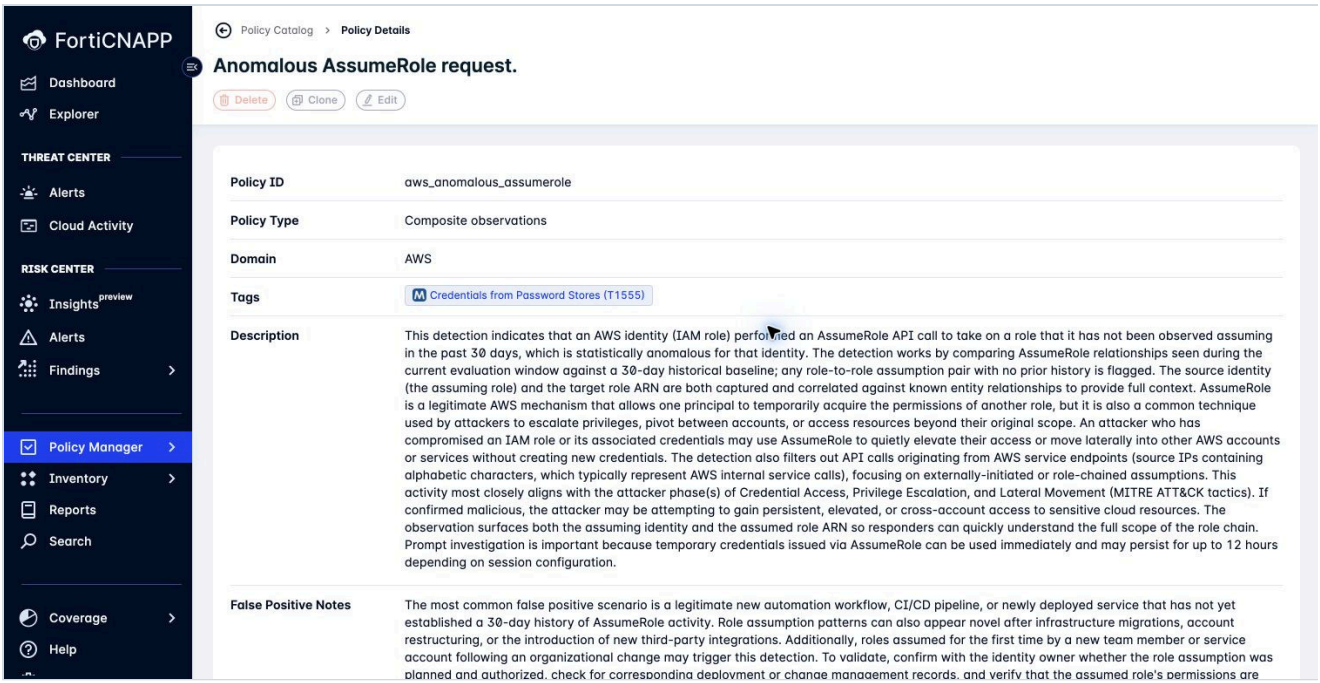
1. Open **Anomalies**.
2. Select **New External Server IP Address Connection From Vulnerable Application**.
3. Read its frequency, severity, domain, and type in **Details**.
4. Open **Query**.
5. Check whether **Clone** is available.



Checkpoint: Your row records **No filter defined** for this example and the state of **Clone**.

Step 7: Read a threat observation definition

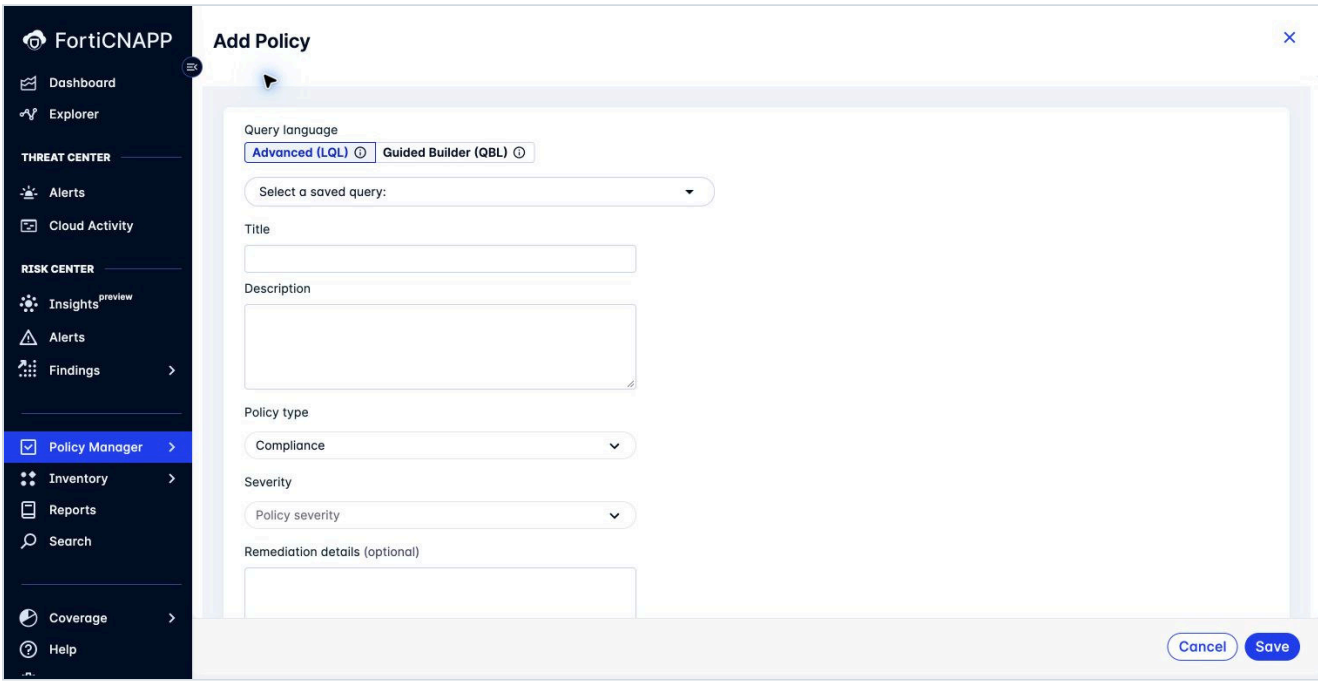
1. Open **Threat Observations**.
2. Select **Anomalous AssumeRole request.** if available. Otherwise, select another definition.
3. Read its description, false positive notes, and triage guidance.
4. Inspect **Edit** and **Clone** without changing the definition.



Checkpoint: Record one triage check and whether the definition can be edited or cloned.

Step 8: Inspect custom policy choices

1. Return to **Risk: Compliance**.
2. Click **Add policy**.
3. Compare **Advanced (LQL)** with **Guided Builder (QBL)**.
4. Inspect **Policy type**.
5. Close the form without saving.



LQL is the query language used in the advanced editor. QBL uses the guided query builder. The LQL path offers Compliance and Violation. The QBL path fixes the policy type to Compliance. A new QBL policy creates a corresponding Risk Insight within about six hours.

Checkpoint: Record the policy types available for each authoring path.

Step 9: Choose a tuning action

Choose one policy from your table. Propose a change using the questions below.

Action	Question to answer first
Change severity	Does business impact justify a different priority?
Change query criteria	Which resources should match, and which should not?
Add an exception	What exact scope and expiry are justified?
Change alert routing	Which team needs this alert?
Disable a policy	What detection or assessment coverage would be lost?

> **What matters:** Choose a control that addresses the problem. Changing severity changes priority; it does not change which resources match.

Checkpoint: Your proposal names an available control, its scope, and the expected result. Include one check for evidence that the change could hide. Keep existing policies unchanged. You will create your own policy in Lab 9.

Next: Lab 9: Build and test a custom query and policy.

Lab 9: Build and test a custom query and policy

Environment: FORTINETAPACDEMO. **Access:** Read/write.

Objectives

Test the condition before you save a policy. Compare the QBL and LQL authoring paths. Create an isolated policy that does not change existing detections.

Concepts

A query selects data that matches a condition. A policy uses that query to define an assessment or detection. Test the query first to check its scope and results. A disabled policy lets you review the configuration before activation.

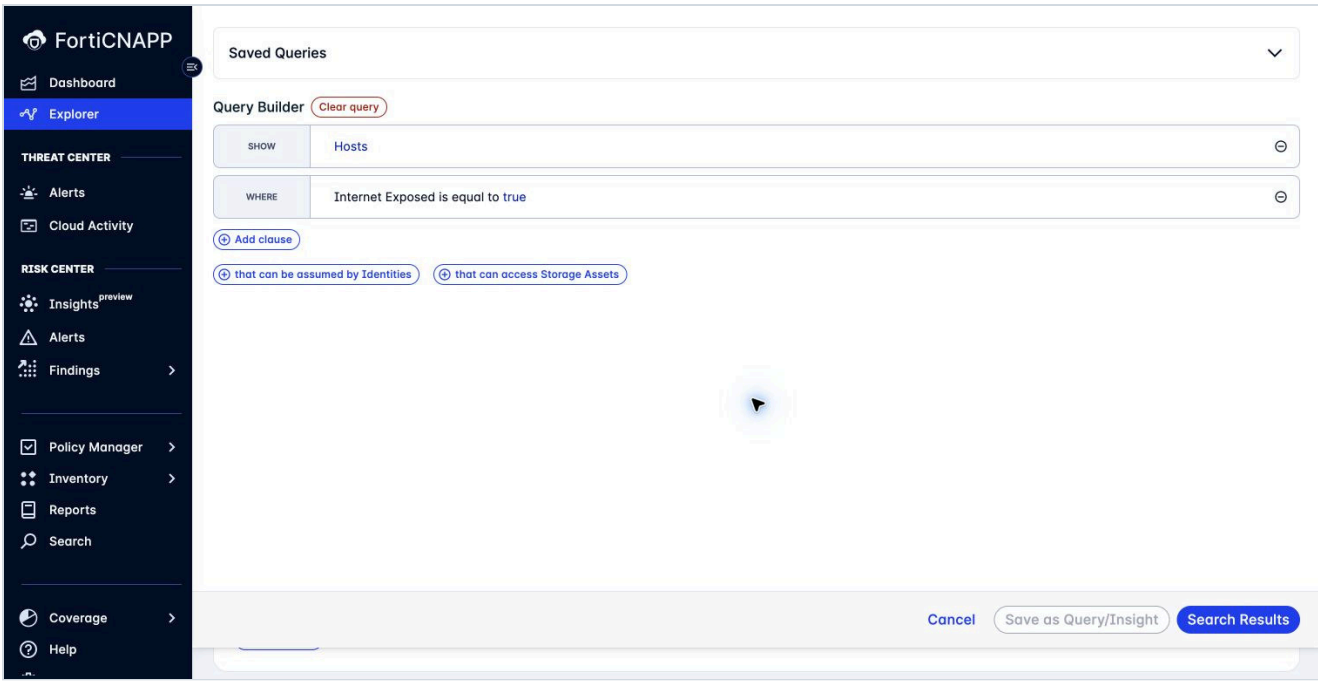
Prerequisites

Complete Lab 8. Use a unique participant ID. Install `lacework` and `jq` using the CLI reference. Clone this repository or download its files. Use the API key supplied for your sandpit account. Screenshots show demo forms. Save objects only in the sandpit.

Lab steps

Step 1: Build a QBL query

1. Confirm **FORTINETAPACDEMO**.
2. Open **Policy Manager > Queries**.
3. Click **Add Query**.
4. Select **Add Explorer Query**.
5. Select **Hosts** as the return type.
6. Click **Add clause** to confirm the return type.
7. Add **Internet Exposed** with the value **True**.
8. Click **Search Results**.
9. Inspect one matching resource.



Checkpoint: Your query contains **Internet Exposed = True**. Open one returned resource and identify the exposure relationship in its graph. If the result is empty, inspect data coverage before concluding that no exposure exists.

Step 2: Save the query

1. Click **Save as Query/Insight**.
2. Set the name to `Workshop-YOURID-exposed-hosts`.
3. Record the generated **Query ID**.
4. Add a description of the intended result.
5. Click **Save Query Only**.
6. Find the query in **Policy Manager > Queries**.

Checkpoint: Your saved query has a unique name and recorded ID. Saving a query does not by itself enable a detection policy.

Step 3: Prepare the sandpit CLI profile

Run these commands from the repository root:

```
lacework version
jq --version
```

Both commands must print a version. The examples use bash or zsh. Replace the key file path with your private path:

```
lacework configure --profile fortinetapacdemo \
  --account partner-demo --subaccount fortinetapacdemo \
  --json_file /PRIVATE/PATH/sandpit-key.json --noninteractive

lacework configure show account --profile fortinetapacdemo
lacework configure show subaccount --profile fortinetapacdemo
```

Checkpoint: The account is `partner-demo`. The subaccount is `fortinetapacdemo`.

> **Caution**
>
> Keep key files and CLI configuration private. Do not paste them into chat, tickets, or Git.

Step 4: Test an LQL condition

Your task is to find EC2 instances that do not require IMDSv2 tokens. Test the condition before saving a disabled policy.

The example finds EC2 configuration where `MetadataOptions.HttpTokens` is not `required`. It checks one IMDSv2 setting, not a complete CIS control.

From the repository root, run:

```
set -o pipefail
lacework query run -f examples/ec2-imdsv2.json \
  --start -7d --end now --limit 5 \
  --profile fortinetapacdemo --account partner-demo \
  --subaccount fortinetapacdemo --json --noninteractive \
  | jq '{sampleRows:length,fields:((.[0] // {}) | keys)}'
```

Checkpoint: The command succeeds. The result reports sample rows and their field names. The five-row limit gives a sample, not a tenant-wide count.

> **Hint:** The seven-day window can include older configuration snapshots. If the result is empty, compare the window with the resource collection time in Explorer.

Step 5: Create a disabled LQL policy

1. Copy `examples/ec2-imdsv2.json` to a private working folder.
2. Replace `REPLACE_ID` in `queryId` with your participant ID.
3. Create the query. Replace `YOUR_QUERY_FILE` with your file path:

```
lacework query create -f YOUR_QUERY_FILE \
  --profile fortinetapacdemo --account partner-demo \
  --subaccount fortinetapacdemo --json --noninteractive
```

4. Record the returned query ID.
5. Prepare a policy file using this shape. Set `queryId` to the exact ID you recorded:

```
title: Workshop-YOURID-IMDSv2
enabled: false
policyType: Violation
alertEnabled: false
queryId: Workshop_IMDSv2_YOURID
severity: high
description: Identify collected EC2 instances that do not require IMDSv2 tokens.
remediation: Review application requirements before requiring IMDSv2 tokens.
```

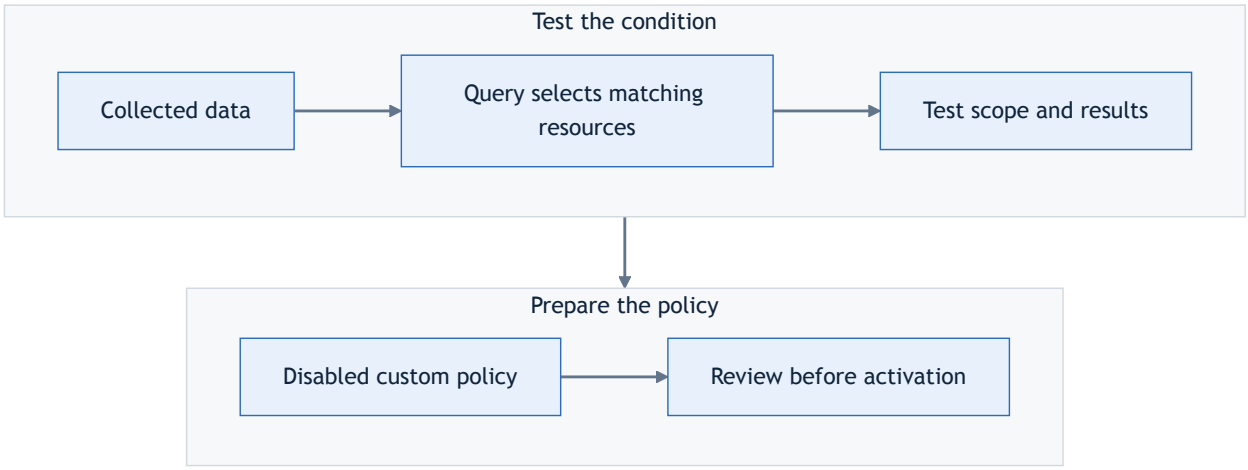
6. Create the policy. Replace `YOUR_POLICY_FILE` with your file path:

```
lacework policy create -f YOUR_POLICY_FILE \
  --profile fortinetapacdemo --account partner-demo \
  --subaccount fortinetapacdemo --json --noninteractive
```

7. Record the returned policy ID.
8. Find the policy in **Policy Manager > Policies**.

9. Confirm that it remains disabled.

Checkpoint: The saved policy references your query. `enabled` and `alertEnabled` are both false.



Step 6: Review a tuning change

Open your own policy. Change its description or severity using the available editor. Keep it disabled. Read the saved value back.

Checkpoint: Read the edited field back from your policy. It matches your change. The policy remains disabled.

Do not enable the policy during this exercise. Activation requires a separate scope and routing review. Remove both queries and the policy in Lab 16.

Next: Lab 10: Build a custom compliance framework.

Lab 10: Build a custom compliance framework

Environment: FORTINETAPACDEMO. **Access:** Read/write.

Objectives

Group policies for a specific reporting purpose. Keep a custom framework distinct from the original CIS benchmark.

Concepts

A framework groups compliance policies into sections for assessment and reporting. A custom framework lets you select controls for a specific review. If you change a CIS framework's policy selection, the result is a custom control set.

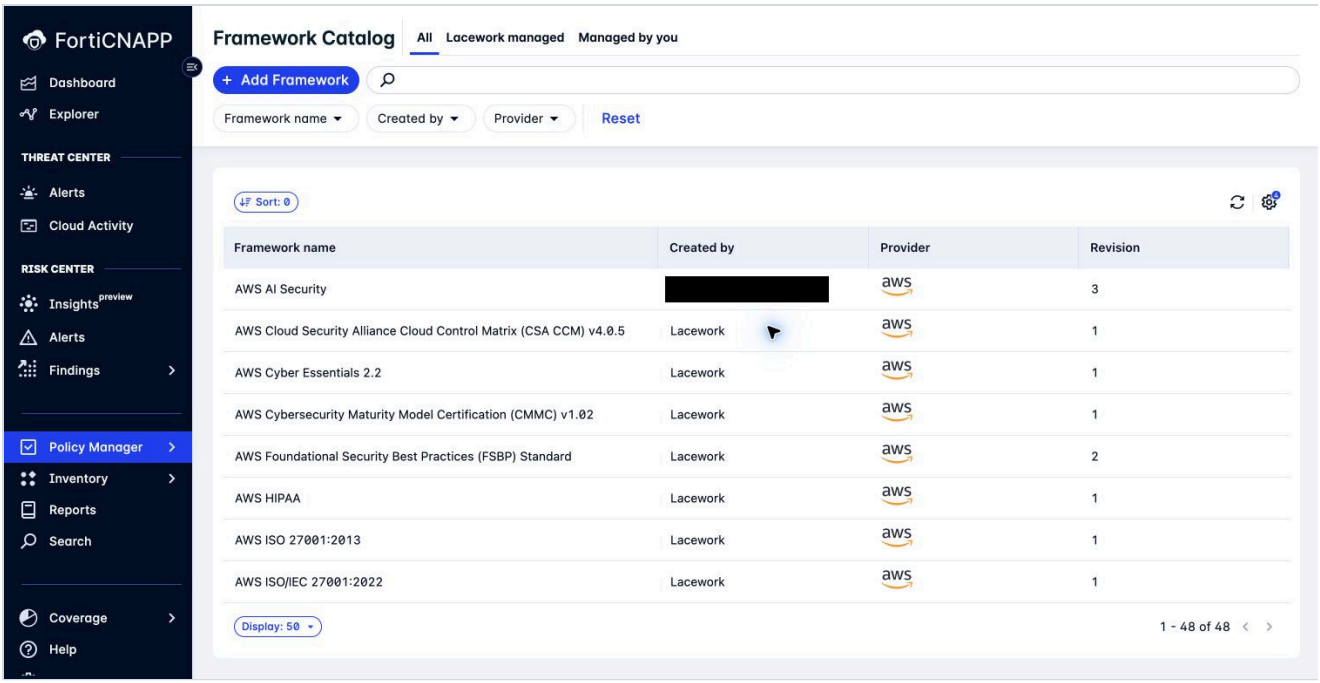
Prerequisites

Complete Lab 7. Use your participant ID. Use the AWS CIS framework that you examined in Lab 7. Screenshots show the demo tenant. Save only in the sandpit.

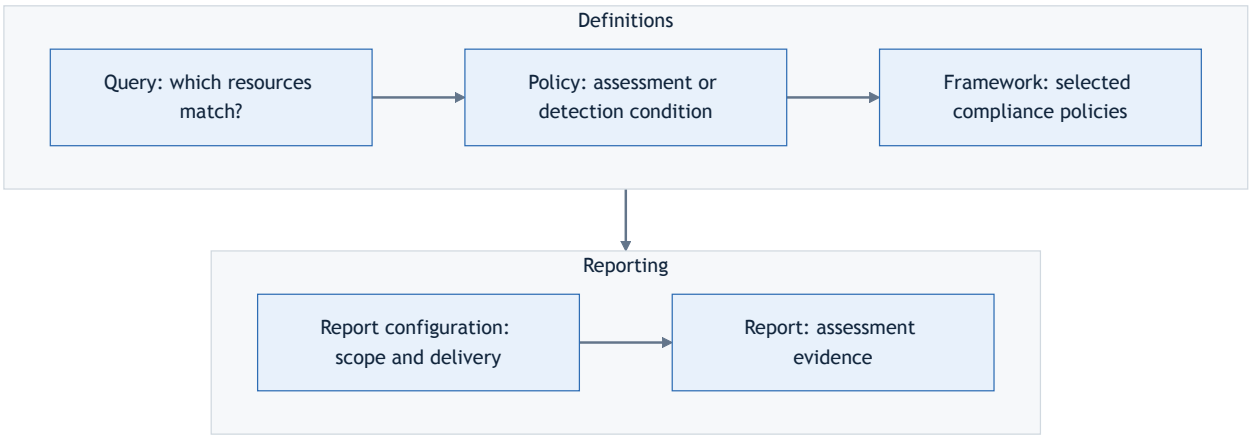
Lab steps

Step 1: Inspect the framework catalog

1. Confirm **FORTINETAPACDEMO**.
2. Open **Policy Manager > Frameworks**.
3. Compare **Lacework managed** with **Managed by you**.
4. Open a built-in framework.
5. Inspect its sections and policy mappings.



Checkpoint: Name one section and a policy mapped to it. The policy defines the check. The framework groups that check with others.



Compliance reporting workflow.

Step 2: Create a custom framework

1. Return to the catalog.
2. Click **Add Framework**.
3. Select the AWS CIS framework from Lab 7.
4. Continue to the framework editor.
5. Set a unique name: `Workshop-YOURID-AWS-review`.
6. Select **Networking** in the copied framework.
7. Click **Add / Edit policies**.
8. Select **Selected policies**.
9. Choose one policy to omit from your custom review.
10. Record why it is outside your review scope.
11. Clear that policy.

12. Click **Update section**.
13. Save your custom framework.

Checkpoint: Your framework appears under **Managed by you**. Reopen its edited section. The omitted policy is absent. The source framework is unchanged.

> **What matters:** A modified CIS-based framework is a custom control set. Do not present it as the complete, unchanged CIS benchmark.

Step 3: Review revision and evaluation state

1. Reopen your framework.
2. Record its revision and policy selection.
3. Check for **Pending changes**.
4. Explain which assessment results may still reflect the earlier configuration.

Checkpoint: Your worksheet records the framework name, provider, revision, and reporting purpose.

Framework changes can take up to 24 hours to appear in assessment results. Continue with the existing assessment while the changes are evaluated. Keep this framework for cleanup. Lab 11 uses the built-in CIS frameworks for its main reports.

Next: Lab 11: Configure AWS and Azure CIS reports.

Lab 11: Configure AWS and Azure CIS reports

Environment: FORTINETAPACDEMO. **Access:** Read/write.

Objectives

Configure separate CIS reports for AWS and Azure. Select the correct framework version, data scope, assessment statuses, and delivery settings.

Concepts

A report configuration selects the framework, resource scope, assessment statuses, and delivery settings. The generated report contains the assessment evidence for those selections. Separate AWS and Azure configurations keep each provider's benchmark and scope clear.

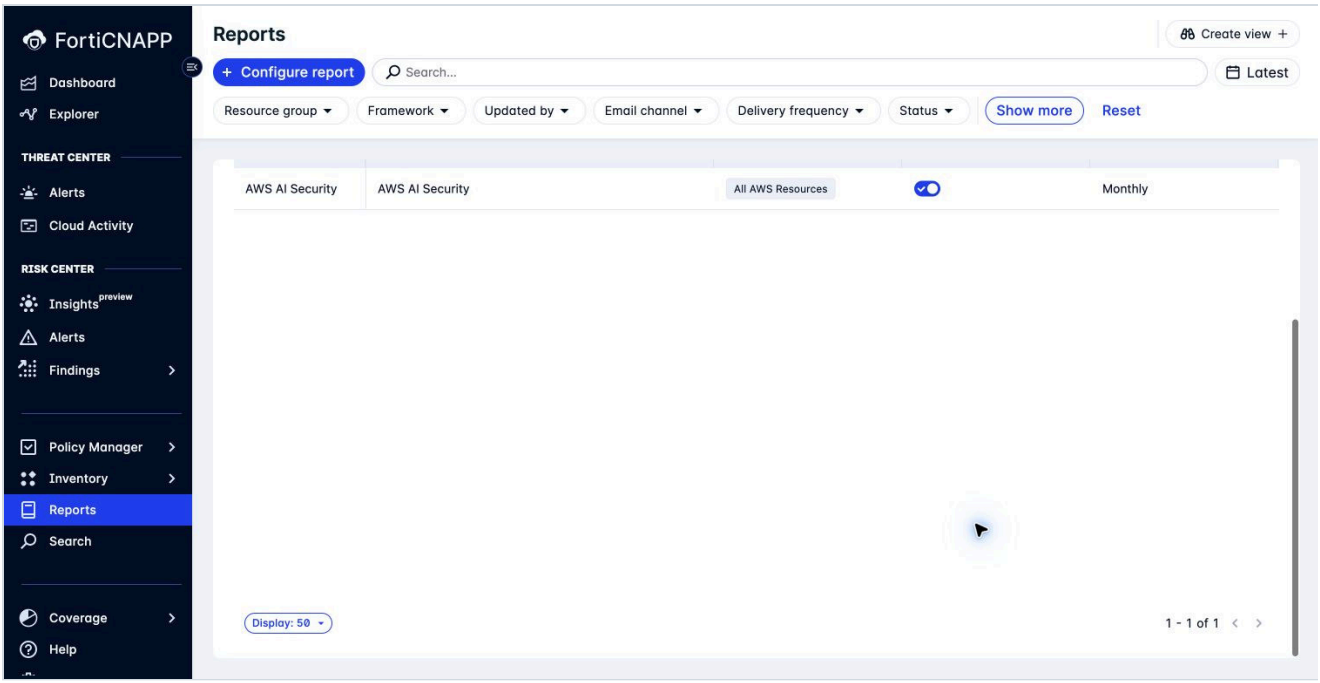
Prerequisites

Complete Labs 7 and 10. Have your assigned AWS and Azure resource groups ready. Screenshots show demo report controls. Create reports only in the sandpit.

Lab steps

Step 1: Inspect existing configurations

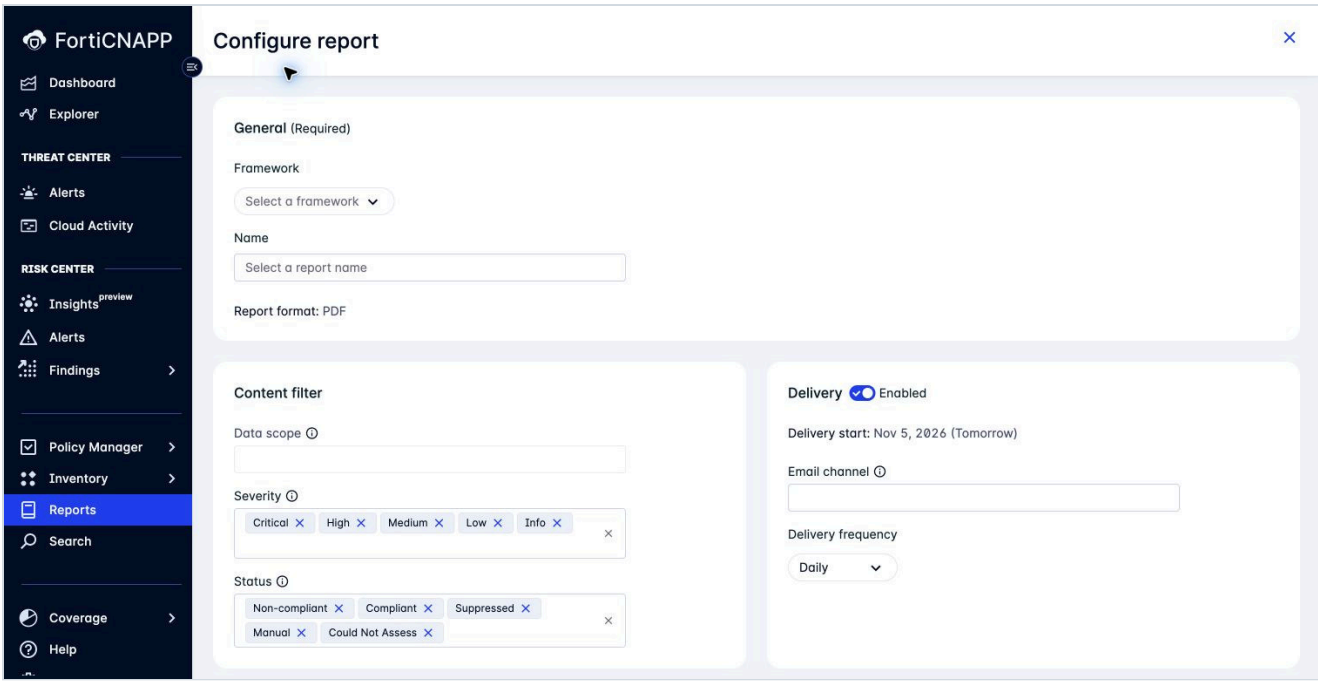
1. Confirm **FORTINETAPACDEMO**.
2. Open **Reports**.
3. Inspect an AWS report configuration.
4. Inspect an Azure report configuration.
5. Record the framework and data scope for each.



Checkpoint: Record the framework and scope from each configuration. These are instructions for report generation, not assessment results.

Step 2: Configure the AWS report

1. Return to **Reports**.
2. Click **Configure report**.
3. Select the required AWS CIS framework and version.
4. Set **Name** to `Workshop-YOURID-AWS-CIS`.
5. Select the approved AWS **Data scope**.
6. Review **Severity**.
7. Review **Status**.
8. Turn off **Enabled** under **Delivery**.
9. Select the **Email channel** assigned for this lab.
10. Review **Delivery frequency**.
11. Create the configuration.
12. Open the saved configuration.
13. Confirm that **Report configuration** is **Disabled**.
14. Turn off its switch if the saved configuration is enabled.



Checkpoint: The saved configuration has the required AWS framework and scope. Delivery is disabled.

An email channel is required even when delivery is off. Check the saved configuration's state. Do not rely only on the creation form's switch.

> **What matters:** A report can hide assessment gaps if you exclude **Manual** and **Could Not Assess** results.

Step 3: Configure the Azure report

Repeat Step 2 with the required Azure CIS framework. Use `Workshop-YOURID-Azure-CIS` as the name. Select the approved Azure subscription scope. Keep delivery disabled.

Checkpoint: Reopen both configurations. Compare their provider, version, and scope with your Lab 7 assessment record. Both remain disabled.

Step 4: Review the delivery plan

1. Record the intended recipients and frequency in your worksheet.
2. Compare them with the selected email channel.
3. Identify the owner who approves enabling delivery.
4. Check the available report preview from **Cloud Compliance > Frameworks**.

Checkpoint: Your delivery plan names the recipients, frequency, framework, and resource scope. Use the saved settings to check each item.

Do not enable delivery to a shared email channel during the lab. Keep both configurations for cleanup in Lab 16.

Next: Lab 12: Check coverage and notification routing.

Lab 12: Check coverage and notification routing

Environment: FORTINETAPACDEMO. **Access:** Read-only review.

Objectives

Check the evidence pipeline before trusting a result. Distinguish a detection policy, an alert route, and a report delivery configuration.

Concepts

Security findings depend on collected data. Missing or stale collection can leave gaps in an investigation. Alert routing sends notifications to a team. Report delivery distributes assessment results. Check collection and delivery separately to find operational gaps.

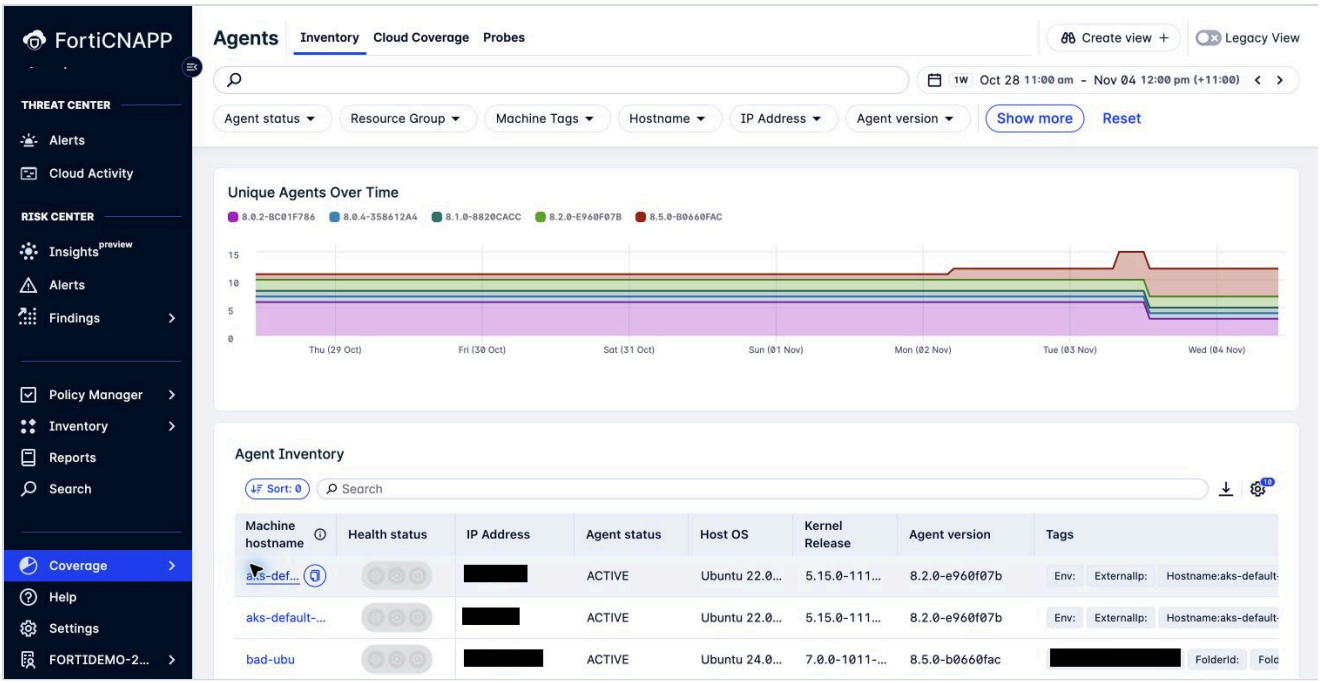
Prerequisites

Complete Lab 11. Do not change shared integrations or channels.

Lab steps

Step 1: Check collection

1. Open **Settings > Cloud accounts**.
2. Inspect the configuration integration for an AWS or Azure scope.
3. Check its state and last successful collection.
4. Inspect the activity log integration separately.
5. Open **Coverage > Agents**.
6. Check agent status and last check-in.
7. Open **Coverage > Agentless**.
8. Compare that coverage with the agent view.



Check collection state in the sandpit.

Checkpoint: Record one collection source and its latest successful update. An enabled integration and a healthy collection state are different checks.

Step 2: Review routing

1. Open **Settings > Channels**.
2. Inspect a channel without changing its recipients.
3. Examine the alert-routing controls available to your role.
4. Trace the intended route for a composite alert.
5. Compare that route with the email channel in your report configuration.

Checkpoint: Record one policy, the alert route you inspected, and the report email channel. Label which defines the condition, routes the alert, and delivers the report.

> **What matters:** Check that alert notifications reach the responsible team.

Step 3: Record an operational issue

Write a short handover for one failed or stale collection. If none is present, use a healthy collection as a status update. Include the integration type, time, observed state, and next check. Keep credentials and unrelated identifiers out of the handover.

Checkpoint: Your handover reports the observed issue without claiming an unverified cause.

Next: Lab 13: Use the CLI with explicit tenant scope.

Lab 13: Use the CLI with explicit tenant scope

Environment: Both tenants. **Access:** Read-only.

Objectives

Retrieve the same evidence through the CLI. Use explicit scope and small JSON summaries. Keep credentials and unrelated tenant details out of shared output.

Concepts

The CLI retrieves FortiCNAPP data through commands that you can repeat or script. A profile stores connection settings and credentials. Account and subaccount flags select the tenant scope. Explicit scope helps you retrieve evidence from the intended environment.

Prerequisites

Install `lacework` and `jq` using the CLI reference. Keep the sandpit profile from Lab 9. Use the API key supplied for your read-only demo account. Clone this repository or download its files. Run commands from the repository root.

Lab steps

Step 1: Check the tools

```
lacework version
jq --version
```

Checkpoint: Both commands print a version. The examples use bash or zsh. Use an approved equivalent shell for other systems.

Step 2: Add the demo profile

Keep the sandpit profile from Lab 9. Add a separate demo profile. Replace the key file path below with your private path.

```
lacework configure --profile fortidemo-2026-04-readonly-use \
  --account partner-demo --subaccount fortidemo-2026-04 \
  --json_file /PRIVATE/PATH/demo-key.json --noninteractive
```

The command saves local credentials. It does not create an API key.

Checkpoint: The configuration command completes without an error. Keep the downloaded key file and local CLI configuration private.

Step 3: Verify the selected scope

```
lacework configure show account --profile fortidemo-2026-04-readonly-use
```

```
lacework configure show subaccount --profile fortidemo-2026-04-readonly-use
```

Checkpoint: The account is `partner-demo`. The subaccount is `fortidemo-2026-04`. Do not print the complete CLI configuration.

Step 4: Find composite alerts

```
set -o pipefail
sh tools/lw-read.sh demo alert list --start -7d --end now \
  | jq '[.[] | select(.derivedFields.category == "Composite")]
    | group_by(.alertName)
    | map({name:.[0].alertName,count:length,
        severities:([.[].severity] | unique)})'
```

Checkpoint: Compare the titles and period with the console. If the selected alert is older, use the same historical period in both tools. Each alert-list request must use a window of seven days or less.

> **Hint:** A month in the console and seven days in the CLI are different datasets. Match time, category, severity, and status before comparing counts.

Step 5: Summarise integration coverage

```
sh tools/lw-read.sh sandpit cloud-account list \
  | jq 'group_by(.type)
    | map({type:.[0].type,total:length,
        enabled:([.[] | select(.enabled == 1)] | length),
        healthy:([.[] | select(.state.ok == true)] | length)})'
```

Compare `enabled` and `healthy` for one integration type. Use **Settings > Cloud accounts** to check the state of its integrations.

Checkpoint: Record the two counts and one integration state from the console. Use that state to explain whether enabled collection is healthy.

The wrapper accepts selected read commands and fixes the tenant flags.

Next: Lab 14: Retrieve evidence through the API.

Lab 14: Retrieve evidence through the API

Environment: Both tenants. **Access:** Read-only.

Objectives

Understand a REST request and its response. Retrieve the alert behind a console investigation. Check time filters, response shape, and pagination before reporting counts.

Concepts

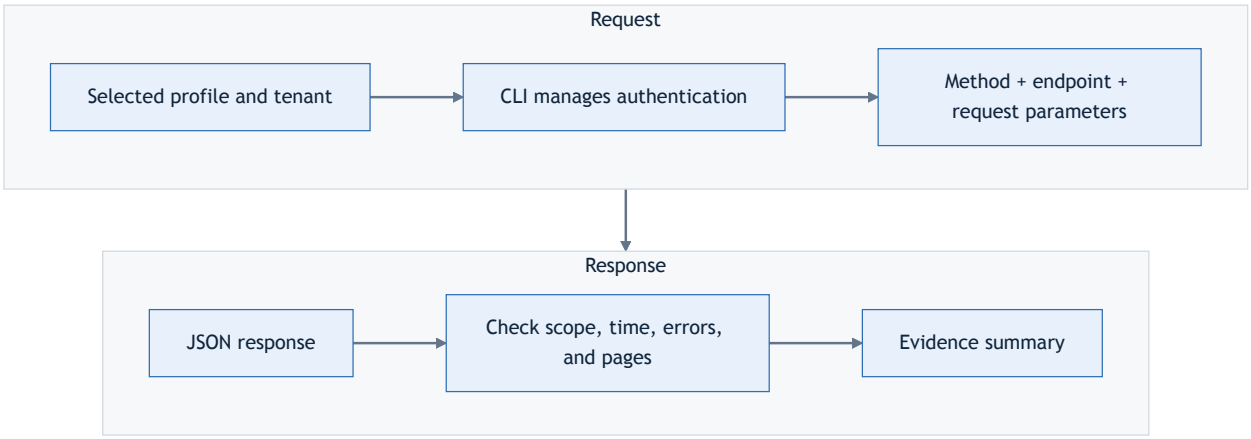
An API request specifies a method, endpoint, and parameters. The response contains data or an error. Filters and pagination determine how much data you receive. Check these details before using a response as evidence or calculating totals.

Prerequisites

Complete Lab 13. Keep an alert ID from your composite investigation. Use the API endpoint documentation for request fields.

Lab steps

Step 1: Identify the request parts



The `lacework api` command manages authentication for the selected profile.

Step 2: Compare a CLI list with an API response

```
set -o pipefail
sh tools/lw-read.sh sandpit api get /api/v2/CloudAccounts \
  | jq '{responseFields:keys,returnedRows:(.data | length)}'
```

Compare the shape with `cloud-account list` from Lab 13. The native list command returns an array. This REST response wraps rows in `data`.

Checkpoint: The API summary reports `responseFields` and `returnedRows`. Identify `data` in the response fields. That is where this response stores its rows.

Step 3: Retrieve one alert

Set the ID from your own investigation:

```
ALERT_ID=REPLACE_WITH_ALERT_ID
ALERT_FILE=$(mktemp)
sh tools/lw-read.sh demo api get \
  "/api/v2/Alerts/${ALERT_ID}?scope=Details" > "$ALERT_FILE"
jq '{responseFields:keys,dataFields:(.data | keys)}' "$ALERT_FILE"
```

Open the response in your private terminal:

```
jq '.data' "$ALERT_FILE" | less
```

In `less`, type `/` followed by an entity value from Lab 2. Press Enter to search. Use `n` for the next match. Press `q` to return to the shell.

1. Find an entity or activity value that also appears in the console.
2. Record the JSON field name and its matching console value in your private worksheet.
3. Repeat for a second value. Record any evidence you could not find.

Checkpoint: Two console values match the API evidence, or your worksheet identifies the missing evidence. Field names alone are not evidence of the incident.

Alert details can contain identities, addresses, and command lines. Keep the file private. Remove it after the comparison:

```
rm -f "$ALERT_FILE"
```

Step 4: Understand search requests

A search endpoint can use `POST` without changing tenant configuration. Check the endpoint documentation to determine its effect.

Run this read-only search in the sandpit:

```
REQUEST_FILE=$(mktemp)
jq -n '{timeFilter:{
  startTime:(now-86400 | floor | todate),
  endTime:(now | floor | todate)}}' > "$REQUEST_FILE"

lacework api post /api/v2/Alerts/search --file "$REQUEST_FILE" \
  --profile fortinetapacdemo --account partner-demo \
  --subaccount fortinetapacdemo --json --noninteractive \
  | jq 'if . == null then {returnedRows:0,hasMore:false}
    else {returnedRows:(.data | length),
      hasMore:(.paging.urls.nextPage != null)} end'

rm -f "$REQUEST_FILE"
```

Check `timeFilter` before you run the request. It selects the last 24 hours.

Checkpoint: The result reports `returnedRows` and `hasMore`. Find the endpoint description that confirms this operation searches alerts.

Step 5: Check completeness

1. Inspect the endpoint's response schema.
2. Check for paging metadata and a next-page link.
3. Retrieve all required pages before calculating a full total.
4. Record the time window and filters with the result.
5. Preserve request failures as failures.

Checkpoint: Record your time window, filters, and `hasMore` value. If another page exists, your current row count is incomplete. A request error is an incomplete check.

> **What matters:** An empty result, a failed request, and a first page are different outcomes. Do not turn any of them into an unsupported claim about tenant security.

Next: Lab 15: Investigate with the FortiCNAPP agent skill.

Lab 15: Investigate with the FortiCNAPP agent skill

Environment: Both tenants. **Access:** Read-only.

Objectives

Use the agent skill to repeat a known investigation. Validate its answer against the underlying evidence. Keep the external agent skill distinct from FortiCNAPP AI Assist and the workload agent.

Concepts

The FortiCNAPP agent skill gives an external coding agent instructions for CLI and API investigations. It can reduce manual evidence collection. You still need to check its sources, scope, and conclusions. It is separate from the workload agent and console AI Assist.

Prerequisites

Complete Labs 2, 13, and 14. Use an approved AI coding agent. Confirm that your organisation permits the selected tenant data in that agent runtime. Keep the CLI profiles from Lab 13.

Lab steps

Step 1: Identify the three agents

Component	Purpose
FortiCNAPP workload agent	Collects workload telemetry
FortiCNAPP AI Assist	Provides assistance inside the console
FortiCNAPP agent skill	Supplies investigation instructions to an external coding agent using the CLI and API

Checkpoint: Explain which component collects process telemetry and which component runs CLI/API investigations.

Step 2: Install or inspect the skill

Use the skill repository for current runtime setup. For Codex, the repository documents this installation target:

```
git clone https://github.com/andrewbearsley/forticnapp-lacework-skill.git \
  ~/.agents/skills/forticnapp-lacework
```

If the skill is already installed, inspect its `SKILL.md` instead. Do not clone over an existing installation. Confirm that the selected runtime loads the skill before continuing.

Checkpoint: Identify the installed skill version and its read-only defaults.

Step 3: Repeat the composite investigation

Invoke `$forticnapp-lacework` in Codex. Supply this task with your chosen ID and period:

```
Use profile fortidemo-2026-04-readonly-use, account partner-demo,
and subaccount fortidemo-2026-04.
Use read operations only. Do not change any tenant state.
Investigate composite alert REPLACE_ID for REPLACE_PERIOD.
Confirm the tenant before requesting data.
Summarise observations, entities, and the activity sequence.
Separate observed facts from hypotheses.
State the evidence sources and any incomplete checks.
Reduce output before displaying it. Keep secrets and unrelated identifiers private.
```

Checkpoint: Compare the result with your Lab 2 worksheet. Check at least two claims against the console or API evidence.

Step 4: Check sandpit coverage

```
Use profile fortinetapacdemo, account partner-demo,
and subaccount fortinetapacdemo.
Use read operations only.
Check cloud integration status and workload agent coverage.
Report counts, stale collection, and failures.
Distinguish data collection health from security findings.
Do not change policies, alerts, integrations, or reports.
```

Checkpoint: A failed integration is reported as an issue. A request failure remains an incomplete check.

Step 5: Review the answer

Ask the agent which source supports its main conclusion. Compare the requested time period with the period it actually queried. Check whether it handled all required pages. Correct any unsupported statement in your worksheet.

Checkpoint: Name one useful result from the agent and one claim that you checked yourself. Cite the console or API evidence used for that check.

Next: Lab 16: Complete the investigation and remove lab objects.

Lab 16: Complete the investigation and remove lab objects

Environment: Demo for investigation; sandpit for cleanup.

Objectives

Present an evidence-based investigation. Confirm report scope and remove your workshop objects.

Concepts

An investigation needs conclusions that others can trace to evidence. Keep detected activity, possible exposure, and compliance results distinct when you present your findings. Remove your saved lab objects to leave the shared sandpit ready for the next exercise.

Prerequisites

Keep your evidence worksheet and the IDs of every object that you created.

Lab steps

Step 1: Present the investigation

1. Confirm **FORTIDEMO-2026-04**.
2. Present the composite alert and its activity window.
3. Explain two related observations.
4. State the affected entities and evidence sources.
5. Separate confirmed facts from hypotheses.
6. Propose a response and a verification method.

Checkpoint: Another participant can trace each main conclusion to its evidence. Do not change the demo alert's status or comments.

Extension: Investigate **Potentially Compromised AWS Credentials** if available. Apply the same method to identities, API activity, and related resources.

Step 2: Explain the risk and compliance evidence

1. Explain an exposure path and a possible interruption.
2. State why that path does not prove exploitation.
3. Identify the CIS framework provider and version.
4. Explain one failed control and one assessment gap.
5. Show the scope of each sandpit report configuration.

Checkpoint: Your explanation keeps incident evidence, exposure, and compliance assessment distinct.

Step 3: Remove your report configurations

1. Switch to **FORTINETAPACDEMO**.
2. Open **Reports**.
3. Find each configuration from your created-object record.
4. Confirm its exact name and framework.
5. Remove only your workshop configurations using the available action.
6. Verify that they no longer appear.

Checkpoint: No workshop report configuration remains for your participant ID.

Step 4: Remove your policies, framework, and queries

1. Open **Policy Manager > Policies**.
2. Remove your disabled workshop policy by its recorded ID.
3. Open **Policy Manager > Frameworks**.
4. Remove your custom workshop framework.
5. Open **Policy Manager > Queries**.
6. Remove your QBL and LQL workshop queries.
7. Verify that each recorded ID is absent.

Remove dependent objects first. Do not remove a query that an existing shared policy uses. Never select all objects for cleanup.

Checkpoint: Your created-object record shows the removal of every object you created.

Step 5: Close the session

Close private exports and credential files. Remove temporary credential files that you no longer need. Do not revoke a shared key or remove another participant's local profile.

Checkpoint: Your workshop objects and temporary credential files are removed.

Return to the workshop overview.

Investigation worksheet

Keep your completed worksheet in approved private storage.

Field	Your evidence
Participant ID	
Tenant	
Alert title and ID	
Time range and timezone	
Composite observations	
Primary entities	
Confirmed facts	
Hypotheses that need checking	
Proposed response	
Attack path entry point	
Target resource	
Proposed path interruption	
CIS provider and benchmark version	
Failed control and affected resource	
Manual or unassessed controls	
Report scope and delivery settings	
CLI/API source and pagination status	
Skill conclusion checked against evidence	

CIS assessments

Provider	Framework and version	Resource scope	Assessment time	Gap to investigate
AWS				
Azure				

Policy review

Category	Example policy	Available controls	Review finding
Compliance			
Vuln Build time			
Vuln Run time			
Data			
Anomalies			
Threat Observations			
Composite Alerts			

Created objects

Tenant	Object type	Name	ID	Removed
FORTINETAPACDEMO				

Completion

Explain the incident evidence to another participant. Explain the difference between an observed action and a possible attack path. Show the CIS framework version and report scope. Confirm that your workshop objects are removed.